

# Diversity and functional specialization of oyster immune cells uncovered by integrative single cell level investigations

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## eLife Assessment

This manuscript offers an exploration of the immune cells in the oyster *Crassostrea gigas*, by correlating distinct hemocyte morphotypes with specific single-cell transcriptional profiles. The evidence supporting the conclusion is **convincing**, deriving from the comprehensive dataset that not only captures unicellular diversity but also associates these cells with distinct immune roles, making it an **important** resource for the broader research community. There are some concerns on the data presentation that leave some questions.

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## Abstract

Mollusks are a major component of animal biodiversity and play a critical role in ecosystems and global food security. The Pacific oyster, *Crassostrea (Magallana) gigas*, is the most farmed bivalve mollusk in the world and is becoming a model species for invertebrate biology. Despite the extensive research on hemocytes, the immune cells of bivalves, their characterization remains elusive. Here we were able to extensively characterize the diverse hemocytes and identified at least seven functionally distinct cell types and three hematopoietic lineages. A combination of single-cell RNA sequencing, quantitative cytology, cell sorting, functional assays and pseudo-time analyses was used to deliver a comprehensive view of the distinct hemocyte types. This integrative analysis enabled us to reconcile molecular and cellular data and identify distinct cell types performing specialized immune functions, such as phagocytosis, reactive oxygen species production, copper accumulation, and expression of antimicrobial peptides. This study emphasized the need for more in depth studies of cellular immunity in mollusks and non-model invertebrates and set the ground for further comparative immunology studies at the cellular level.

## Introduction

*Mollusca* is the second largest invertebrate phylum, after *Arthropoda*, and the largest marine phylum, comprising approximately 23 % of all known marine organisms (1). Among them, bivalves exhibit a high diversity and a rich evolutionary history (2). The Pacific oyster *Crassostrea (Magallana) gigas* (*C. gigas* - Thunberg, 1793) (NCBI:txid29159) is a sessile filter-feeding bivalve that thrives in a variety of stressful environments ranging from intertidal to deep-sea conditions (3). It is a key species for the aquaculture industry worldwide (4).

Several infectious diseases affect *C. gigas* at different life stages, which impacts its production. Given the significant socio-economic value of this species, there has been an increased focus on understanding and mitigating these diseases (5). The causes of these mortalities can involve a variety of pathogens, including viruses, bacteria and parasites that can be responsible for the mortality events affecting *C. gigas* (6). One of the most extensively researched infectious diseases is POMS (Pacific Oyster Mortality Syndrome), a polymicrobial disease responsible for mass mortalities of juvenile oysters (7). The disease is triggered by the OsHV-1  $\mu$ Var herpesvirus, which alters the immune defenses of oysters, allowing the colonization of opportunistic bacteria, including *Vibrio*, that cause hemocyte lysis and bacteremia, ultimately leading to animal death (7). Other bacterial pathogens have also been identified as a contributing factor in mass mortalities of adult Pacific oysters in several countries. The most notable is *Vibrio aestuarianus* which affects adult oysters in Europe (8, 9). To date, the majority of oyster pathogens or opportunistic pathogens that have been characterized in detail have been found to subvert hemocyte defenses for their own benefit. These include the OsHV-1  $\mu$ Var herpesvirus (10) and virulent *Vibrio* strains of the species *Vibrio crassostreae* and *Vibrio tasmaniensis* (11), *Vibrio aestuarianus* and *Vibrio harveyi* (12), highlighting the critical role of hemocytes in oyster immunity. The development of immune- based prophylactic treatments, such as immune-priming and immune-shaping, represents a promising avenue for enhancing the natural defenses of oysters against pathogens and increasing their survival rate. Nevertheless, the advancement of such therapies is still constrained by a dearth of knowledge regarding the underlying molecular and cellular mechanisms (13, 14).

The study of hemocytes has a long history, dating back to the 1970s (for review see (15)). Hemocytes are cellular effectors of the immune system. They engage in phagocytosis to engulf and destroy potential pathogens, neutralizing parasites by encapsulation, or preventing pathogen dissemination by cell aggregation and the release of extracellular DNA traps (16).

Furthermore, they engage in the humoral response by releasing cytokines, antimicrobial peptides, and reactive oxygen species (ROS), which enable them to combat pathogens (17). In addition to their role in oyster immunity, hemocytes have been implicated in numerous physiological processes, including shell repair (18), wound healing, nutrient transport, and environmental contaminant removal (19). Despite this acquired knowledge, hemocytes remain an under-characterized population of circulating immune cells. The lack of a unified classification and of molecular and functional genetic tools hinders our understanding of lineage ontogeny and functional specialization. Several studies have proposed different classifications of hemocytes in the *Ostreidae* family, with 3 to 4 hemocyte types reported (15). These classifications are primarily based on either microscopic or flow cytometry analyses. In *C. gigas*, three primary hemocyte cell types have been classically identified : blast, hyalinocyte, and granulocyte cells. Blast-like cells are considered as undifferentiated hemocyte types (20), hyalinocytes (21) seem to be more involved in wound repair, and granulocytes, more implicated in immune surveillance. The latter are considered as the main immunocompetent hemocyte types (22). While the immune response of *C. gigas* has been extensively studied using classical transcriptomics (23–25) at the whole animal, tissue, or circulating hemocyte levels, these

approaches have failed to consider the diversity of hemocyte cell types and lineages that underpin these responses (26). However, it is still imperative to accurately describe the diversity of these cells, understand their ontogeny, and delineate cell lineages to comprehend their specific roles. The advent of single-cell RNA sequencing (scRNA-seq) techniques has enabled the monitoring of global gene expression at the single-cell level with thousands of individual cells in a single experiment. This provides a unique opportunity to overcome these limitations and deepen our understanding of hemocyte diversity and function in bivalves. Recently scRNA-seq was used to provide a first molecular description of a hemocyte population in the oyster *Crassostrea hongkongensis* (27). However, in the absence of morphological and/or functional characterization studies, the authors could not deduce the hemocyte cell types to be matched to the transcriptomic profiles generated by scRNA-seq.

While numerous transcriptomic analyses have been conducted on *C. gigas* hemocytes, none have adopted a single-cell approach. In this study, we present an integrative analysis of the diversity of *C. gigas* hemocytes at the single-cell level. To this end, we combined scRNA-seq from a pathogen-free adult oyster combined with cytological, cell fractionation and functional assays. This approach allowed us to create a comprehensive transcriptomic, cytological and functional atlas of hemocyte cell types. Our scRNA-seq analyses identified 7 distinct transcriptomic populations and functional annotation revealed distinct populations with specific functions, including phagocytosis, oxidative burst, energetic metabolism, enhanced transcription, translation and cell division. Quantitative cytology enabled the identification of 7 morphologically distinct hemocyte cell types, which allowed us to reconcile molecular and cytological data. Density gradients were used to separate hemocyte cell types and qPCR or functional assay analyses were performed to validate cell type-specific markers. By employing this integrated approach, we could identify 1 type of hyalinocyte, 2 types of blasts and 4 types of granular cells. Furthermore, we identified cell types that perform antimicrobial functions through phagocytosis, ROS production, copper accumulation, and expression of antimicrobial peptides. Finally, trajectory analysis of scRNA-seq data combined with functional analysis revealed distinct differentiation pathways that may control hemocyte ontology and differentiation processes. Based on these findings, we propose a more comprehensive and up-to-date classification of *C. gigas* hemocytes, with a more accurate description of the different cell types, their potential ontology and a precise description of their sub-functionalization.

## Results

### Single-cell RNA sequencing reveals 7 distinct transcriptomic clusters of circulating immune cells in oysters

Oysters are known to exhibit a high degree of individual genetic polymorphism, including Copy Number Variation (CNV) and Presence Absence Variation (PAV) (28). To prevent misinterpretation of the single-cell transcriptomic data, and to characterize the hemocyte cell types and their heterogeneity, we sampled hemocytes from a unique known pathogen-free animal (Ifremer Standardized Animal, 18-month-old) and applied single-cell drop-seq technology. A total of 4,950 cells were loaded, with a target recovery of 3,000 single hemocytes for analysis. (Fig. 1A). The scRNA-seq library was generated and sequenced, resulting in 127,959,215 high-quality filtered reads available for single-cell analysis (ENA project accession number PRJEB74031). Primary bioinformatics analysis was performed using the STAR solo aligner software (29) against the *C. gigas* genome (Genbank reference GCA\_902806645.1) from the Roslin Institute (30). Of the 127,959,215 reads, 97 % showed a valid barcode and 89.2 % were successfully mapped to the genome with a saturation of 75.6 %. A total of 2,937 cells were profiled, yielding a median of 1,578 genes and 4,412 unique molecular identifiers (UMIs) per cell among the 23,841 total genes detected, with a sequencing saturation of 75.6 % (Supp. Table S1). Secondary bioinformatic data processing was conducted using the Seurat R package (version 4.3.0) (31).

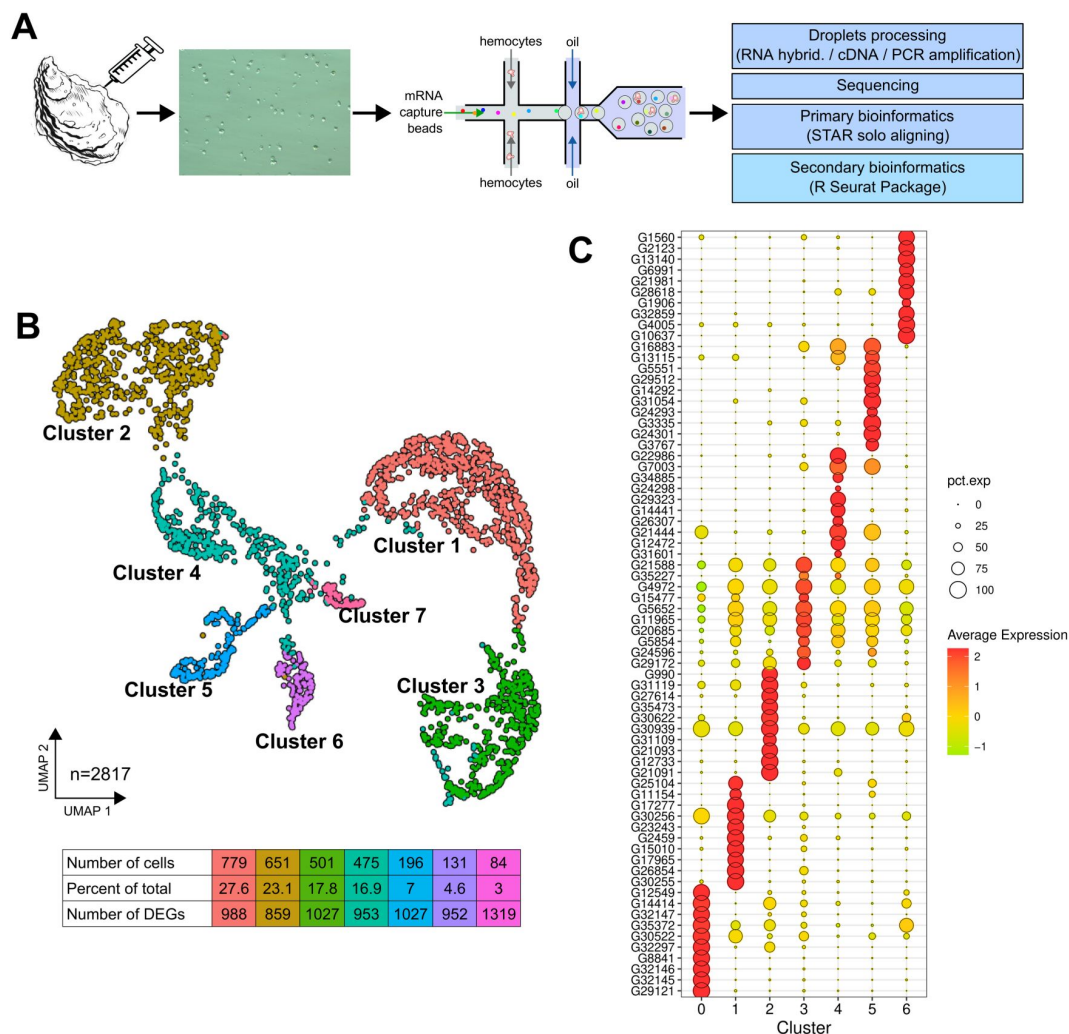
The data set was filtered to remove data corresponding to empty droplets or cell doublets. Cells with a gene number between 750 and 4,000 and less than 5 % mitochondrial genes were retained. After quality control processing, 120 cells corresponding to empty droplets and cell doublets were removed, and 2,817 cells were processed for data normalization. Finally, we performed linear dimensional reduction and clustering on the 3,000 most variable genes from 2,817 cells (Supp. Fig. S1). Dimension reduction and clustering led to identifying 7 transcriptomic clusters, within which hemocytes were distributed. These 7 different clusters represented 27.6, 23.1, 17.8, 16.9, 7, 4.6 and 3 % of the total cells (**Fig. 1B**). For each transcriptomic cluster, the heatmap displays the top 10 positively enriched marker genes identified by Seurat's differential expression analysis, highlighting cluster-specific overexpression (**Fig. 1C**).

Average Log2FC values and percentage of expression in each cluster relative to all other clusters were calculated for the ten most differentially expressed genes (**Fig. 1D** and **Table 1**). Clear transcriptomic cell clusters were detected as well as specific gene markers for each cluster (Supp. Data 1). Only the transcriptomic signature of cluster 4 was less contrasting than that of the other clusters (**Fig. 1D**).

## KEGG pathways and GO-terms analyses reveal functional diversity in *C. gigas* hemocytes

The ScRNA-seq data demonstrated that specific functions are carried out by the hemocyte cell types that comprise the seven transcriptomic clusters. A preliminary overview of the functions over-represented in each cluster was obtained through a KEGG pathway analysis on the overrepresented transcripts (Log2FC > 0.25) using the *C. gigas* annotation provided by the DAVID consortium (32), thereby identifying pathways specifically enriched in each cluster. Cluster 1 demonstrated enrichment in viral processing and endocytosis (**Figure 2A** and Supplemental Table S2). Additionally, it demonstrated enrichment in pyruvate metabolism, glycolysis/gluconeogenesis and the pentose phosphate pathways. Clusters 1 and 3 exhibited a distinctive enrichment in carbohydrate metabolism and endocytosis. In particular, cluster 3 exhibited enriched transcripts associated with glycolysis/gluconeogenesis and TCA cycle activities. Cluster 4 was enriched in protein synthesis, including transcription (spliceosome, ribosome, nucleocytoplasmic transport and mRNA surveillance pathway) folding, sorting and degradation of protein pathways. Clusters 2, 5 and 6 exhibited a shared signature of enrichment in ribosome-related genes. Cluster 2 demonstrated a specific enrichment in motor protein-coding genes responsible for cell motility and xenobiotic metabolism. Remarkably, clusters 1, 3, 4 and 5 exhibited enriched oxidative phosphorylation transcripts, whereas the transcripts of cluster 7 were enriched in vesicular trafficking and endo-lysosomal pathways (endocytosis, endosome, phagosome, lysosome, auto and mitophagy). Cluster 7 displayed also the signature of Wnt pathway components.

For a more detailed functional characterization of each transcriptomic cluster, we performed a re-annotation of the *C. gigas* genome using a combination of tools to enhance the GO term richness of the existing annotation prior to functional GO term analysis. The original annotation for *C. gigas*, based on the work of Penaloza et al. (30), provided GO annotations for 18,750 out of 30,724 genes (61%). In order to improve upon these baseline annotations, we used the *C. gigas* genome (Genbank reference GCA\_902806645.1) and the associated gff3 annotation file from the Roslin Institute (30). These files were used to extract and process the longest CDSs for GO-term annotation using the Orson pipeline (see Materials and Methods). Of the 30,724 extracted CDSs, 22,462 were annotated (GO-terms and sequence description), yielding an annotation percentage of 73.1 %. Of the 30,724 CDSs, 22,391 were annotated with Molecular Functions (MF), Biological Processes (BP), and Cellular Components (CC) GO- terms (Supp. Fig. S2 and Supp. Data 2). Overall, this improved annotation coverage significantly enhances the resolution of our downstream functional analyses. Using the GO- term annotation and the Log2FC of genes calculated after



**Fig. 1.**

### scRNA-seq analysis of *C. gigas* circulating hemocytes reveals 7 transcriptomic cell clusters

**(A)** Schematic of the scRNA-seq 10X Genomics Chromium microfluidic technology and bioinformatics processing workflow used. Dissociated hemocytes were collected from a pathogen-free oyster and encapsulated in droplets for processing. After sequencing, the data were processed bioinformatically. **(B)** Uniform Manifold Approximation and Projection (UMAP) plot for dimensional reduction of the data set and summary of cells and the number of Differentially Expressed Genes (DEGs) in each cluster. The table shows the characteristics (number of cells, percentage of total cells and number of Differentially Expressed Genes in each cluster) of the seven clusters identified. **(C)** Dot plot representing the ten most enriched DEGs per cluster based on average expression (avg\_log2FC). The color gradient of the dot represents the expression level, while the size represents the percentage of cells expressing each gene per cluster.

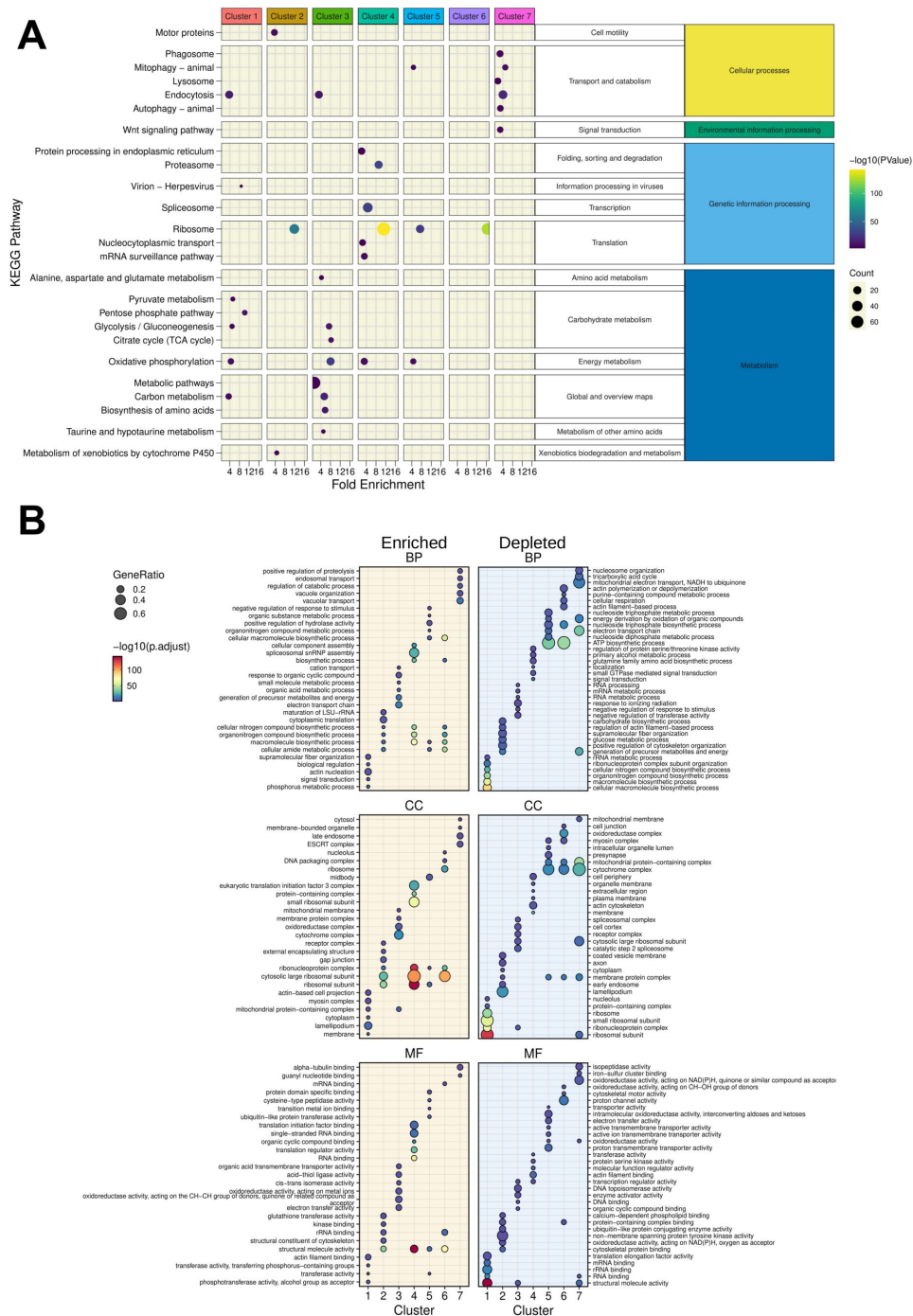


| Gene      | log2FC | Pct 1 | Pct 2 | Description                                                      | Cluster |
|-----------|--------|-------|-------|------------------------------------------------------------------|---------|
| Cluster 1 |        |       |       |                                                                  |         |
| G29121    | 4.31   | 1     | 0.099 | L-galactono-gamma-lactone oxidase                                |         |
| G32145    | 4.18   | 0.992 | 0.079 | Angiotensin-1 receptor                                           |         |
| G32146    | 3.90   | 0.985 | 0.063 | Angiotensin-1 receptor                                           |         |
| G8841     | 3.13   | 0.988 | 0.049 | C-type lectin domain-containing protein                          |         |
| G32297    | 2.88   | 0.99  | 0.232 | PNPLA domain-containing protein                                  |         |
| G30522    | 2.76   | 1     | 0.515 | Cytochrome P450 2D8                                              |         |
| G35372    | 2.69   | 1     | 0.483 | 60S ribosomal protein L26                                        |         |
| G32147    | 2.45   | 0.969 | 0.131 | Angiotensin-1 receptor                                           |         |
| G14414    | 2.45   | 1     | 0.343 | DUF4773 domain-containing protein                                |         |
| G12549    | 2.38   | 0.974 | 0.132 | Glutaredoxin domain-containing protein                           |         |
| Cluster 2 |        |       |       |                                                                  |         |
| G30255    | 5.31   | 1     | 0.15  | Metalloproteinase inhibitor 3                                    |         |
| G26854    | 4.24   | 1     | 0.16  | Stanniocalcin                                                    |         |
| G17965    | 4.21   | 0.94  | 0.04  | Complement C1q-like protein 2                                    |         |
| G15010    | 3.99   | 0.99  | 0.17  | Kyphoscoliosis peptidase                                         |         |
| G2459     | 3.82   | 1     | 0.21  | Uncharacterized protein                                          |         |
| G23243    | 3.51   | 0.98  | 0.07  | Fibronectin type-III domain-containing protein                   |         |
| G30256    | 3.23   | 1     | 0.69  | NTR domain-containing protein                                    |         |
| G17277    | 3.21   | 0.99  | 0.04  | Putative modulator of levamisole receptor-1                      |         |
| G11154    | 3.16   | 0.68  | 0.05  | Collagen alpha-1(XII) chain                                      |         |
| G25104    | 3.1    | 0.84  | 0.08  | C1q domain-containing protein                                    |         |
| Cluster 3 |        |       |       |                                                                  |         |
| G21091    | 5.51   | 0.99  | 0.13  | X-box binding protein-like protein                               |         |
| G12733    | 5.42   | 1     | 0.12  | Galectin                                                         |         |
| G21093    | 5.17   | 0.94  | 0.05  | IrfG                                                             |         |
| G31109    | 4.36   | 0.77  | 0.03  | Cystatin domain-containing protein                               |         |
| G30939    | 4.25   | 1     | 0.96  | Peptide ABC transporter permease                                 |         |
| G30622    | 3.95   | 1     | 0.21  | BHLH domain-containing protein                                   |         |
| G35473    | 3.86   | 0.96  | 0.04  | Uncharacterized protein                                          |         |
| G27614    | 3.78   | 0.96  | 0.13  | G Protein Receptor F1-2 domain-containing protein                |         |
| G31119    | 3.77   | 0.99  | 0.42  | Arrestin C domain-containing protein                             |         |
| G990      | 3.6    | 0.93  | 0.03  | LITAF domain-containing protein                                  |         |
| Cluster 4 |        |       |       |                                                                  |         |
| G29172    | 3.09   | 0.78  | 0.40  | G Protein Receptor F1-2 domain-containing protein                |         |
| G24596    | 2.37   | 0.78  | 0.25  | High mobility group protein DSP1                                 |         |
| G5854     | 1.8    | 0.64  | 0.38  | N-acetyltransferase domain-containing protein                    |         |
| G20685    | 1.59   | 0.88  | 0.53  | Ribosomal protein L22                                            |         |
| G11965    | 1.57   | 0.92  | 0.7   | Ribosome L4 associated C domain-containing protein               |         |
| G5652     | 1.54   | 0.97  | 0.75  | 60S ribosomal protein L30                                        |         |
| G15477    | 1.52   | 0.55  | 0.3   | Metalloendopeptidase                                             |         |
| G4972     | 1.49   | 0.96  | 0.8   | 60S ribosomal protein L12                                        |         |
| G35227    | 1.46   | 0.55  | 0.06  | ML domain-containing protein                                     |         |
| G21588    | 1.42   | 0.94  | 0.69  | 60S ribosomal protein L13                                        |         |
| Cluster 5 |        |       |       |                                                                  |         |
| G31601    | 5.24   | 0.38  | 0.06  | Polyribonucleotide nucleotidyltransferase                        |         |
| G12472    | 3.99   | 0.85  | 0.03  | Galectin                                                         |         |
| G21444    | 3.58   | 1     | 0.41  | Uncharacterized protein                                          |         |
| G26307    | 3.42   | 0.6   | 0.02  | Uncharacterized protein                                          |         |
| G14441    | 3.35   | 0.84  | 0.08  | CUBN                                                             |         |
| G29323    | 3.21   | 0.9   | 0.01  | GTPase IMAP family member 4                                      |         |
| G24298    | 3.18   | 0.29  | 0.05  | Putative Transmembrane protease serine 9                         |         |
| G34885    | 3.13   | 0.59  | 0.01  | Gliding motility-associated C-terminal domain-containing protein |         |
| G7003     | 2.55   | 0.99  | 0.16  | ETS-related transcription factor Elf-4                           |         |
| G22986    | 2.37   | 0.94  | 0.1   | Metalloendopeptidase                                             |         |
| Cluster 6 |        |       |       |                                                                  |         |
| G3767     | 6.45   | 0.74  | 0.02  | Natterin-1                                                       |         |
| G24301    | 6.16   | 0.99  | 0.02  | Aspartate-semialdehyde dehydrogenase                             |         |
| G3335     | 5.19   | 1     | 0.16  | ncRNA                                                            |         |
| G24293    | 4.57   | 0.59  | 0.01  | Uncharacterized protein                                          |         |
| G31054    | 3.56   | 1     | 0.13  | GATA-binding factor 3                                            |         |
| G14292    | 3.29   | 0.87  | 0.03  | Membrane-associated guanylate kinase inverted 3                  |         |
| G29512    | 3.2    | 0.99  | 0     | Caveolin                                                         |         |
| G5551     | 2.93   | 0.97  | 0.02  | Neuronal acetylcholine receptor subunit alpha-6                  |         |
| G13115    | 2.73   | 0.83  | 0.26  | ncRNA                                                            |         |
| G16883    | 2.62   | 1     | 0.2   | Allograft inflammatory factor                                    |         |
| Cluster 7 |        |       |       |                                                                  |         |
| G10637    | 4.21   | 0.99  | 0.03  | BZIP domain-containing protein                                   |         |
| G4005     | 4.07   | 0.99  | 0.21  | Ras-related protein Rab-20                                       |         |
| G32859    | 3.9    | 0.93  | 0.04  | G Protein Receptor F1-2 domain-containing protein                |         |
| G1906     | 3.62   | 0.5   | 0.02  | Glutathione peroxidase                                           |         |
| G28618    | 3.53   | 0.91  | 0.09  | Sporozoite and liver stage asparagine-rich protein               |         |
| G21981    | 3.5    | 0.94  | 0.05  | ncRNA                                                            |         |
| G6991     | 3.4    | 0.85  | 0.01  | Rho-GAP domain-containing protein                                |         |
| G13140    | 3.12   | 1     | 0.03  | DBH-like monooxygenase protein 1                                 |         |
| G2123     | 3.05   | 0.94  | 0.02  | SPRY domain-containing SOCS box protein 3                        |         |
| G1560     | 3.02   | 0.95  | 0.17  | Glutathione peroxidase                                           |         |

**Table 1.**

## Top 10 overexpressed genes identified in each transcriptomic cluster.

The first column indicates the gene number according to the annotation. 'log2FC' represents the log2 fold change of the gene in the cluster compared to all other cells. 'Pct1' is the percentage of cells expressing the gene in the cluster and 'Pct2' is the fraction of cells expressing the gene in all other clusters. The description is the annotation of the expressed gene (adjusted p-value < 0.05).



**Fig. 2.**

## KEGG and Gene Ontology analysis of the gene signature in each cluster.

(A) A synthetic representation of the KEGG pathway analysis is shown. Colored columns represent the 7 transcriptomic clusters. Each row is a KEGG pathway, the colored dot represents the  $-\log(p\text{-value})$  and the dot size represents the number (count) of enriched genes in each pathway category. The fold enrichment is shown on the x-axis. Panels (B) show the results of Gene Ontology terms (GO-terms) for Biological Processes (BP), Cellular Components (CC) and Molecular Functions (MF) respectively, obtained with the genes of each cluster (Absolute value  $\text{Log2FC} > 0.25$  and significant  $p\text{-value} < 0.001$ ) using RBGOA analysis ( $p\text{-value} \leq 0.001$ ) for three different ontology universes. Each panel corresponds to one ontology universe, and the analysis highlights enriched and depleted terms. The dot size indicate the proportion of significant GO-terms and the gradient scale the  $p\text{-adjusted}$  value.

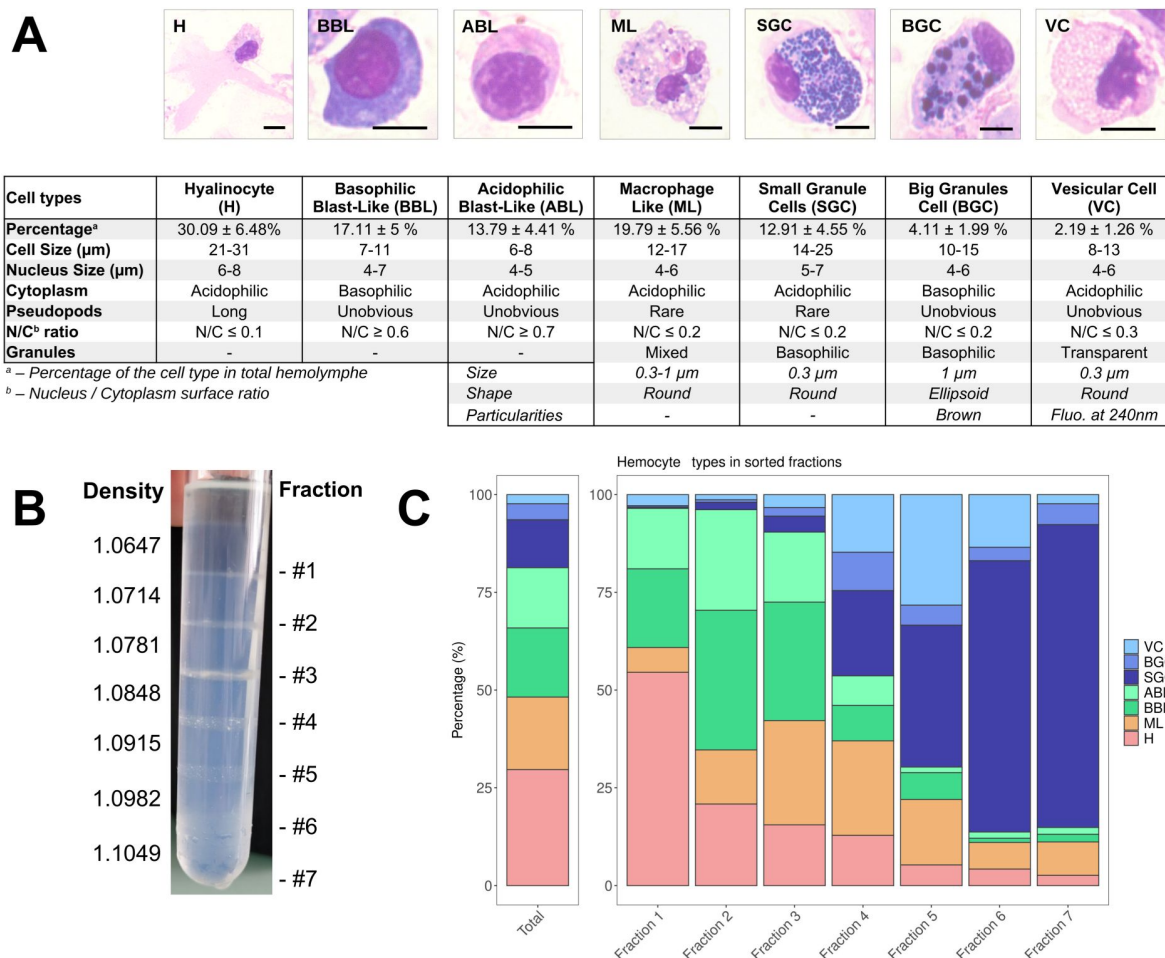
scRNA-seq processing in each cluster, GO enrichment analysis was performed by rank-based gene ontology analysis (RBGOA) (33 [↗](#)). RBGOA analysis was performed on the GO-terms identified in each cluster (Supp. Data 3). The results are presented in **Figures 2B, C and D** [↗](#).

The scRNA-seq-based analysis identified seven distinct transcriptomic profiles for each cell cluster, thereby shedding light on greater heterogeneity and functional diversity of *C. gigas* hemocytes than previously described. Cluster 1, comprising 27.6 % of cells, is characterized by GO-terms related to myosin complex, lamellipodium, membrane and actin cytoskeleton remodelling, as well as phosphotransferase activity. This is evidenced by an enrichment in actin nucleation, phosphorus metabolic process and signal transduction BP, actin filament binding and phosphotransferase activity MF and lamellipodium and actin-based cell projection CC. Cluster 2, comprising 23.1 % of cells, has an increased translation activity, as indicated by enrichment in BP, MF and CC terms related to rRNA binding, cytoplasmic translation, maturation of LSU-rRNA, and ribosomes respectively. Cluster 3, representing 17.8 % of cells, shows enrichment in cellular oxidation and actin nucleation, as evidenced by an enrichment in BP and MF related to oxidoreductase activities acting on metal ions, electron transfer and cellular oxidation, and CC related to cytochromes, oxidoreductase complex, and mitochondrial protein-containing complexes. Cluster 4, representing 16.9 % of cells, has transcriptomic signatures reminiscent of spliceosome assembly, RNA maturation, and macromolecules biosynthesis (BP related to biosynthetic process, macromolecules biosynthesis process, and spliceosome assembly, MF related to RNA and ssRNA binding and translation initiation, and CC related to ribosomes and ribonucleoprotein complexes). Cluster 5, comprising 7 % of cells, demonstrates enrichment in structural molecule activity, hydrolase regulation and ubiquitin-like protein transferase activity, as evidenced by its BP, MF, and CC related to ribosome and midbody localization. Cluster 6, comprising 4.6 % of cells, is characterized by a specific nucleosome organization, translation, and biosynthetic processes, with related BP, MF, and CC tied to nucleolus, ribosomes, ribonucleoprotein complex and rRNA binding. Cluster 7, representing a mere 3 % of cells, is characterized by multivesicular body sorting, vacuolar transport, vacuole organization, and positive regulation of proteolysis, with related BP, MF, and CC of the ESCRT machinery.

## Seven morphologically distinct immune cell types are identified by quantitative cytology and transcriptomic markers

Cytological studies were conducted using MCDH staining to better characterize the diversity of circulating hemocytes in *C. gigas*. Seven distinct hemocyte morphotypes were identified: three non-granular (acidophilic blasts, basophilic blasts and hyalinocytes) and four granular (small granule cells, big granule cells, vesicular cells and macrophage-like) (**Fig. 3** [↗](#) and Supp. Fig. S3). Hyalinocytes (30 % of total hemocytes) are large cells with an irregular spreading membrane. They contain an azurophilic cytoplasm without granulations and their irregular nucleus varies in size (**Fig. 3A** [↗](#) panel H). Basophilic blasts characterized by a basophil cytoplasm (**Fig. 3A** [↗](#) panel BBL) and acidophilic blasts with acidophil cytoplasm (**Fig. 3A** [↗](#) panel ABL) accounted for 17 % and 14 %, respectively, of the total hemocytes. They are rounded cells without granulation, with a uniform and regular dense nucleus and a high nucleo-cytoplasmic ratio. Macrophage-like cells (20 % of the hemocytes) present an irregular membrane punctuated by rare pseudopodia with a polylobed nucleus and a basophilic cytoplasm where polychromatic inclusions of various sizes could be observed (**Fig. 3A** [↗](#) panel ML). Small granule cells (13 % of total hemocytes) have an irregular membrane punctuated by rare pseudopodia, an acidophilic cytoplasm, and numerous homogeneous purple granules (**Fig. 3A** [↗](#) panel SGC). Big granule cells (4 % of total hemocytes) are rounded cells with a basophilic cytoplasm containing large, slightly dark purple to black vesicles of heterogeneous size (**Fig. 3A** [↗](#) panel BGC). Vesicular cells (2 % of total hemocytes) are rounded cells with acidophilic cytoplasm, rich in homogeneous transparent and fluorescent to UV light vesicles with an irregular nucleus (**Fig. 3A** [↗](#) panel VC). Only blasts (BBL, ABL) have a nucleo-cytoplasmic ratio greater than 0.6, whereas the other hemocytes described have a nucleo-cytoplasmic ratio less than 0.3.





**Fig 3.**

### **C. gigas** naive hemocyte formula and Percoll gradient hemolymph fractionation

**(A)** Morphology, percentages and characteristics of the 7 cell types identified by MCDH staining. **H** : Hyalinocyte, **ML** : Macrophage Like, **BBL** : Basophilic Blast Like cell, **ABL** : Acidophilic Blast Like cell, **SGC** : Small Granule Cell, **BGC** : Big Granule Cell, **VC** : Vesicular Cell. Scale bar : 5 μm. For the ML, SGC, BGC, and VC hemocyte types, size refers to the average granule diameter, shape describes the morphology of the granules (e.g., round, ellipsoid), and particularities highlight distinguishing features such as granule color or fluorescence properties observed under specific staining or imaging conditions. **(B)** Sorting of hemocytes on a discontinuous Percoll gradient. 7 fractions were identified along the gradient at the top of each density cushion (from d=1.0647 at #1 to d= 1.1049 at #7). **(C)** Representation of the average values (from 5 different fractionation experiments) of the different hemocyte types in the seven percoll gradient fractions compared to the average hemolymph composition of a naive oyster (Total). **VC** : Vesicular Cells, **BGC** : Big Granule Cells, **SGC** : Small Granule Cells, **ABL** : Acidophilic Blast Like cells, **BBL** : Basophilic Blast Like cells, **ML** : Macrophage Like cells and **H** : Hyalinocytes respectively. (Supp. Fig. S4 for statistics).

To gain further insight into the functional characterization of these diverse hemocyte morphotypes and to associate them with the transcriptomic clusters identified in the scRNA-seq, we enhanced an existing hemocyte fractionation approach using an isopycnic Percoll density gradient to sort the hemocytes (34). Cell sorting was performed using a discontinuous Percoll gradient with densities ranging from 1.0647 to 1.1049. Seven distinct density fractions were established (Fig. 3B) and the hemocyte composition was then characterized by cytological analysis of each fraction (Fig. 3C; see also Supp. Fig. S4 for statistical significance). The uneven distribution of hemocyte morphotypes along the density gradient enabled a relative separation of the different cell populations (Supp. Fig S4). In summary, hyalinocytes were significantly enriched in the first fraction, while macrophage-like cells were significantly enriched in fraction 3 and fraction 4. Acidophilic blasts were significantly enriched in fraction 2. Basophilic blasts were significantly enriched in fractions 2 and 3.

Vesicular cells were enriched in fraction 5, big granule cells were enriched in fraction 4 and small granule cells in fractions 6 and 7 (Fig. 3C and Supp. Fig. S4).

To identify the transcriptomic cell clusters corresponding to the different hemocyte morphotypes, we used RT-qPCR to detect the expression of different cluster-specific marker genes in the different hemocytes in the Percoll density fractions. The marker genes were selected without any a priori assumptions based solely on their enrichment levels (Log2FC) and their percentage of expression (Pct1 / Pct2 ratio) in the cluster of interest relative to all other cell clusters (Table 2). The expression level of each marker in the seven different clusters confirmed that the 14 selected marker genes were differentially expressed in each scRNA-seq transcriptomic cluster (Supp. Fig. S5).

RT-qPCR expression profiles in the hemocyte fractions obtained from the Percoll density gradient revealed distinct patterns according to the different transcriptomic cluster marker genes (Fig. 4A). Cluster 1 markers (LACC24 and CLEC) were overexpressed in fractions 1, 2, 3 and underexpressed in the remaining fractions relative to total hemocytes. Cluster 2 markers (EGFL and LEVAR) showed a decreasing pattern of expression from fraction 1 to fraction 7. Cluster 3 markers (TGC and XBOX) showed a significant increase in expression in fractions 4 to 7. Cluster 4 markers (MLDP and HMGB1) showed a gradually decreasing expression pattern from fraction 2 to fraction 7. Cluster 5 marker genes (GAL and CUBN) were underexpressed in fractions 4 to 7, but not differentially expressed in fractions 1 to 3 compared to total hemocytes. Cluster 6 marker genes (CAV and NAT1) were overexpressed in fraction 3, expressed similarly to total hemocytes in fraction 2, and underexpressed in fractions 1, and 4 to 7. Finally, cluster 7 marker genes (MOX and GPROT) were overexpressed only in fractions 3 to 7. Correlation analysis and statistical validation demonstrated a clear association between hemocyte morphotypes and 14 gene markers (Fig. 4B, Fig. 4C and Supp. Table S4). Cluster 3 marker genes (TGC and XBOX) correlated positively ( $r = 0.74$  &  $0.75$ ) and specifically with small granule cells (SGC). Cluster 7 markers (MOX and GPROT) correlated ( $r = 0.68$  and  $r = 0.56$ ) with vesicular cells (VC). Cluster 2 markers (EGFL and LEVAR) correlated positively with hyalinocytes (H) ( $r=0.85$  and  $r=0.81$ ). No specific markers could be identified for blasts and macrophage-like cells.

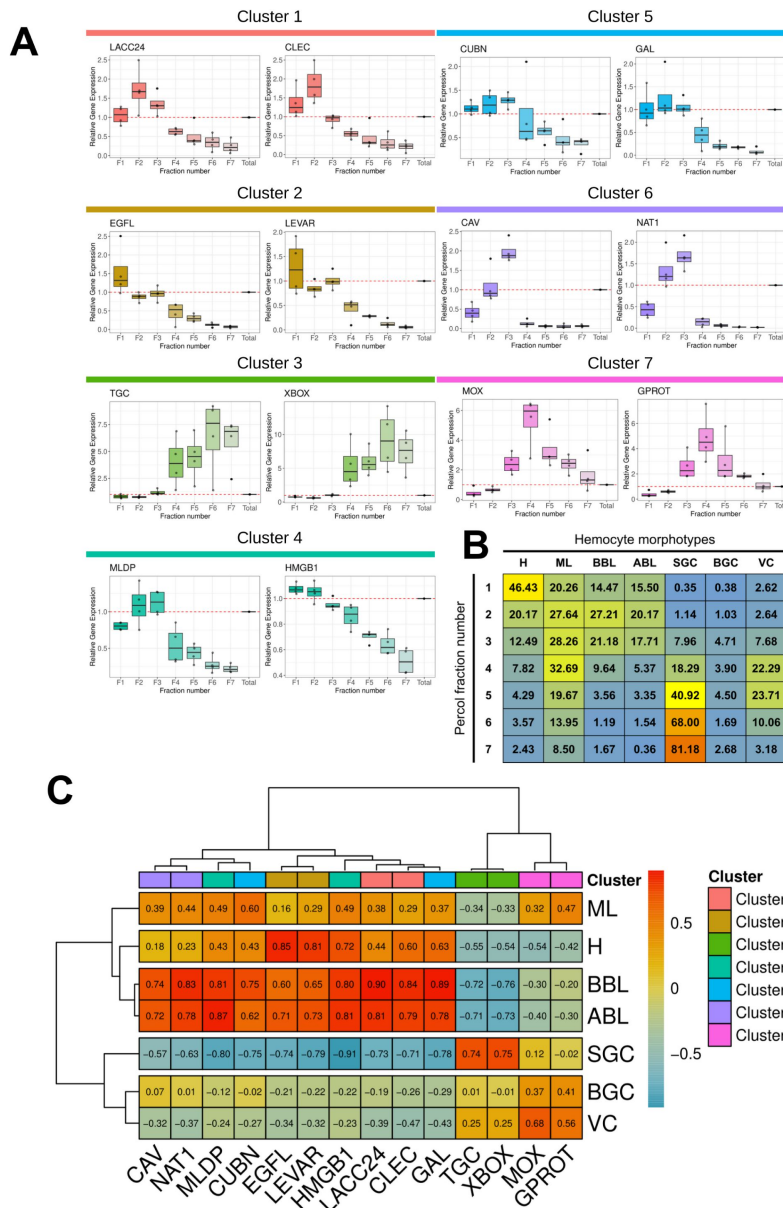
By employing RT-qPCR, cell sorting on Percoll gradients, and scRNA-seq analysis, we could identify cell types corresponding to three transcriptomic clusters. Cluster 3 corresponds to small granule cells (SGC), cluster 2 to hyalinocytes (H), and cluster 7 to vesicular cells (VC). Cluster 3, cluster 2 and cluster 7 represent 17.8 %, 23 % and 3 % of the total cells analyzed by scRNA-seq, respectively. This is consistent with cytological data, which indicated the presence of 12.5 %  $\pm$  5 % Small Granule Cells, 30 %  $\pm$  9.3 % Hyalinocytes and 2.3 %  $\pm$  1.6 % Vesicular Cells (Fig. 1B and Fig. 3A). Furthermore, the molecular and cellular functions derived from GO-term and KEGG analyses (Fig. 2) were in alignment with the anticipated functions for these three cell types.

| Gene Number | Avg. Log2FC | Pct.1/Pct.2 Ratio | Cluster | Description                                       | Marker Name |
|-------------|-------------|-------------------|---------|---------------------------------------------------|-------------|
| G8994       | 2.05        | 38.20             | 1       | Laccase-24                                        | LACC24      |
| G8841       | 3.13        | 20.16             |         | C-type lectin domain-containing protein           | CLEC        |
| G24846      | 2.31        | 31.52             | 2       | EGF-like domain-containing protein 8              | EGFL        |
| G17277      | 3.21        | 23.00             |         | Putative modulator of levamisole receptor-1       | LEVAR       |
| G22387      | 3.16        | 16.76             | 3       | TGc domain-containing protein                     | TGC         |
| G21091      | 5.51        | 7.69              |         | X-box binding protein-like protein                | XBOX        |
| G35227      | 1.46        | 9.98              | 4       | ML domain-containing protein                      | MLDP        |
| G31074      | 0.94        | 6.51              |         | High mobility group protein B1                    | HMGB1       |
| G14441      | 3.35        | 11.01             | 5       | Cubilin                                           | CUBN        |
| G12472      | 3.99        | 24.91             |         | Galectin                                          | GAL         |
| G29512      | 3.20        | 246.25            | 6       | Caveolin                                          | CAV         |
| G3767       | 6.45        | 41.11             |         | Natterin-1                                        | NAT1        |
| G13140      | 3.12        | 40.00             | 7       | DBH-like monooxygenase protein 1                  | MOX         |
| G32859      | 3.90        | 23.82             |         | G protein receptor F1-2 domain-containing protein | GPROT       |

**Table 2.**

**Table of the 14 marker genes specific to the different transcriptomic clusters.**

Gene number, average Log2FC, pct1/pct2 ratio (percentage of cells expressing this transcript in the cluster divided by the percentage of all other cells expressing this transcript) and cluster number are reported. The description is taken from our annotation and the marker name is derived from the description.



**Fig 4.**

### Characterization of molecular markers specific to the different hemocyte morphotypes.

(A) Relative gene expression level of the 14 markers in the various fractions after gradient density sorting. The graphs show the relative expression of genes compared to their expression in total hemocytes in the various fractions (red dotted line). Relative gene expression levels were normalized to the reference gene *Cg-rps6* and the  $2^{-\Delta Ct}$  method was used to calculate relative expression levels, where  $\Delta Ct$  represents the difference between the target gene's Ct value and the reference gene's Ct value. Standard deviations were calculated based on four independent experiments. (B) Average percentage of each hemocyte type in the 7 Percoll gradient fractions used to quantify marker gene expression by qPCR. (C) Correlation matrix between the relative gene expression of each marker gene in each fraction and the percentage of each hemocyte type in the same fractions. Values and color scale represent the Pearson correlation coefficient ( $r$ ) ranging from -1 (inverse correlation) to +1 (full correlation). **H** : Hyalinocyte, **ML** : Macrophage Like, **BBL** : Basophilic Blast Like cell, **ABL** : Acidophilic Blast Like cell, **SGC** : Small Granule Cell, **BGC** : Big Granule Cell, **VC** : Vesicular Cell. **LACC24** : Laccase 24, **CLEC** : C-type lectin domain-containing protein, **EGFL** : EGF-like domain-containing protein 8, **LEVAR** : Putative regulator of levamisole receptor-1, **TGC** : TGC domain-containing protein, **XBOX** : X-box binding protein-like protein, **MLDP** : ML domain containing protein, **HMGB1** : High mobility group protein B1, **CUBN** : Cubilin, **GAL** : Galectin, **CAV** : Caveolin, **NAT1** : Natterin-1, **MOX** : DBH-like monooxygenase protein 1, **GPROT** : G protein receptor F1-2 domain-containing protein.

## Only macrophage-like and small granule cells behave as professional phagocytes

The hemocytes separated on the Percoll gradient were characterized functionally to gain further insight into their functional specialization. Two hemocyte functions, namely phagocytosis and production of Reactive Oxygen Species (ROS), are known to carry out major cellular antimicrobial activities. Granular cells have been suggested to be the professional phagocytes specialized for these functions (22 [↗](#)). Phagocytosis and oxidative burst were studied using cell response toward zymosan particles (35 [↗](#)).

The phagocytic activity of hemocytes was first tested on a sample of total oyster hemolymph. Cells were incubated for 1 hour with either zymosan particles or the bacterial strain *Vibrio tasmaniensis* LMG20012<sup>T</sup>. Only the small granule cells and the macrophage-like cells exhibited efficient phagocytosis for both zymosan or vibrios, as observed after MCDH staining (Fig. 5A [↗](#), Supp. Fig. 6 and Supp. Fig. 7A). Macrophage-like cells and small granule cells showed a phagocytic activity of 49 % and 55 %, respectively, and a phagocytosis index of 3.5 and 5.2 particles per cell respectively (Fig. 5B [↗](#) and Supp. Fig. 7B), as confirmed in 3 independent experiments. Very limited phagocytic activity was observed for hyalinocytes (1.7 %), basophilic blasts (0.18 %), and big granule cells (2.7 %) with a phagocytosis index of 2, 1, and 2.5 particles per cell, respectively (Fig. 5B [↗](#) and Supp. Fig. 7B). These results confirmed that only small granule cells and macrophage-like cells behave as professional phagocytes demonstrating robust phagocytic activity.

## Only macrophage-like cells produce Reactive Oxygen Species

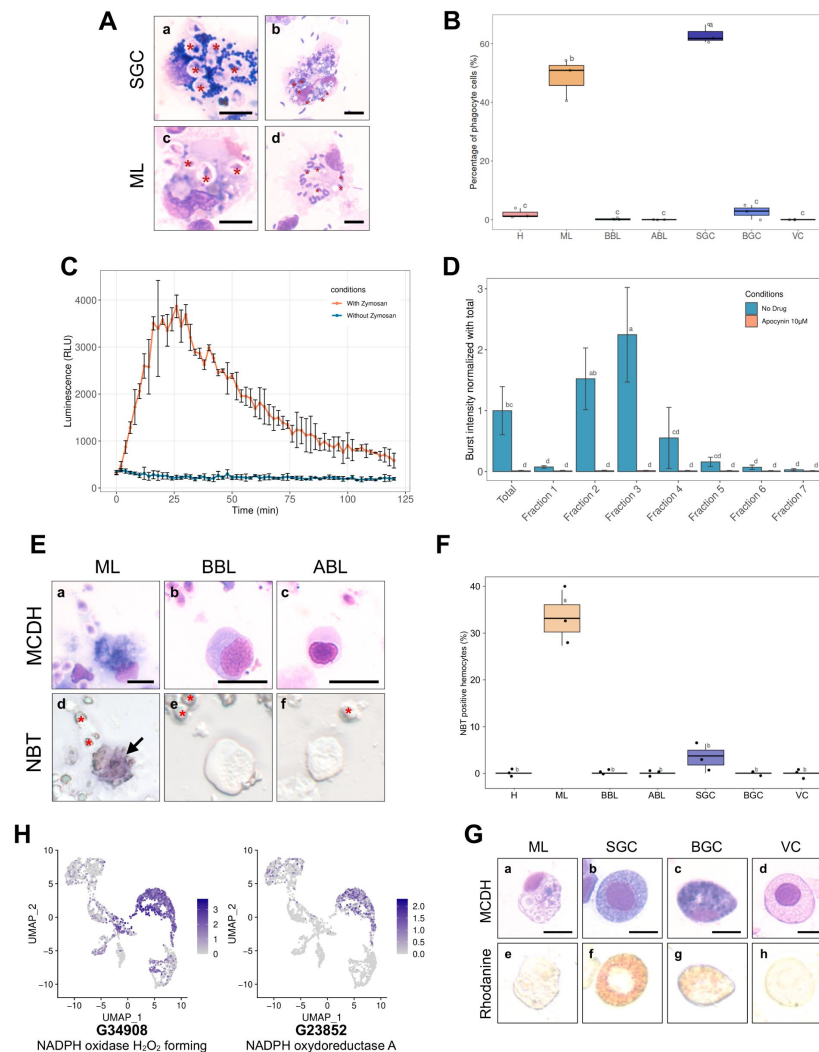
The next step was to assess the capacity of *C. gigas* hemocytes in each Percoll fraction to produce Reactive Oxygen Species (ROS) upon stimulation by zymosan, utilizing the Luminol oxidation assay. Luminol luminescence peaked 25 minutes after exposure of hemocytes (isolated from total hemolymph) to zymosan, indicating a robust oxidative burst (Fig. 5C [↗](#)).

The production of ROS was dependent on a NADPH oxidase, as it was completely inhibited by the NADPH oxidase-specific inhibitor apocynin (Fig. 5D [↗](#)). To ascertain which hemocyte types were involved in ROS production we tested Percoll density-sorted hemocytes for oxidative burst activity. Fractions 2 and 3 displayed higher oxidative burst activity than the other fractions. The burst intensity of fraction 3 was twice that of total hemocytes, and fraction 2 also exhibited significantly increased burst activity. In contrast, fraction 4 showed a significant decrease in oxidative burst, and no activity was observed for fractions 1, 6 and 7 (Fig. 5D [↗](#)). These results indicate that NADPH-dependent oxidative burst activity is carried out by fractions enriched in macrophage-like (ML) cells and blast-like cells (ABL and BBL).

Previous studies have shown that granular cells can produce ROS (22 [↗](#)). In contrast, we found that small granule cells collected in fraction 7 could not produce ROS through an oxidative burst. To identify which type of hemocyte from blast-like cells and macrophage-like cells produces ROS within a few minutes after exposure to zymosan, ROS production was investigated using NitroBlueTetrazolium (NBT) reduction to stain hemocytes directly.

Correlative microscopic analysis using MCDH staining can then be conducted after the NBT reduction reaction. Macrophage-like cells were strongly and significantly stained by NBT reduction (33.2 %) (Fig. 5E [↗](#), panels a & d). In contrast, some small granule cells were lightly stained by NBT reduction (Fig. 5F [↗](#)). Blast-like cells, big granule cells, vesicular cells, and hyalinocytes were never NBT stained, confirming that these cell types were not involved in ROS production (Supp. Fig. S7). Taken together, these observations demonstrate that small granule cells and macrophage-like cells are the two professional phagocytes among hemocytes. However, only macrophage-like cells are capable of oxidative burst upon exposure to zymosan. In light of these





**Fig 5.**

### Phagocytosis, Reactive Oxygen Species production capacity and copper storage of hemocytes.

(A) Images of small granule cells (SGC) and macrophage-like (ML) cells with phagocytosed zymosan particles (panels a & c - red stars) and *Vibrio tasmaniensis* LMG20012<sup>T</sup> bacteria (panels b & d - red stars) from whole hemolymph sample. Scale bar: 5 µm. (B) Quantification of the phagocytic activity of zymosan particles by each cell type. The graph shows the result of 3 independent experiments. (C) Luminescence recording to detect the production of Reactive Oxygen Species (ROS). In orange, a biphasic curve was obtained on naive oyster hemolymph after zymosan addition at t = 0 min. In blue, the control condition corresponds to hemocytes without zymosan addition. (D) Graph showing the intensity of ROS production in each Percoll fraction. Normalized burst intensity was calculated from the luminescence peak obtained from each fraction. In blue, no drug was added to the experiment, in orange, ROS production was impaired by the addition of apocynin. (E) NBT (NitroBlueTetrazolium) staining of hemocytes exposed to zymosan particles. Hemocytes morphology after MCDH staining: Macrophage Like (a), Basophilic (b) and Acidophilic (c) Blast cells. NBT staining of the different types of hemocytes (d-f). Red stars show zymosan and bacteria particles. Black arrows indicate Macrophage-Like cells. Scale bar: 10 µm (F) Quantification of NBT-positive cells present in the total hemolymph of oysters exposed to zymosan. (H) UMAP plots showing cells expressing NADPH oxidase found in the scRNA-seq dataset and their expression level. (G) Labeling of intracellular copper stores in *C.gigas* hemocytes. MCDH (upper panels) and rhodanine (lower panels) staining of oyster hemocytes to reveal copper accumulation. Scale bar: 10µm. For panels (B), (D) and (F) the alphabetic characters displayed above the data points in each plot represent statistically significant differences among the groups, as determined by Tukey's test following ANOVA. Groups denoted by different letters differ significantly at the p < 0.05 level of statistical significance. H : Hyalinocytes, ABL : Acidophilic Blast-Like cells, BBL : Basophilic Blast-Like cells, ML : Macrophage-Like cells, SGC : Small Granule Cells, VC : Vesicular Cells and BGC : Big Granule Cells.

new functional data, we further analyzed the expression level of NADPH-oxidase-related enzymes in the scRNA-seq dataset. The cells in cluster 1 predominantly expressed two NADPH oxidase isoforms (gene numbers G34908 and G23852) compared to other clusters (**Fig. 5H** [↗](#)). Furthermore, this cluster expresses macrophage-related genes, including the macrophage-expressed gene 1 protein (G30226) (Supp. Data S1), along with maturation factors for dual oxidase, an enzyme involved in peroxide formation (Supp. Fig. S8), supporting its designation as macrophage-like based on functional characteristics. Collectively, these data indicated that cells of transcriptomic cluster 1 corresponded to macrophage-like cells.

## Small granule cells and big granule cells accumulate intracellular copper

The effects of copper on oyster hemocytes have been studied extensively due to its abundance in polluted marine environments ([19](#) [↗](#)). In addition, several studies have shown that *Vibrio* species pathogenic for *C. gigas* possess copper resistance genes, which are crucial for their survival within hemocytes ([9](#) [↗](#)). This prompted us to investigate which hemocyte types are involved in copper metabolism. To this end, total hemocytes were isolated from naive oysters and stained with rhodanine to reveal copper storage in cells. Rhodanine staining revealed that 33 %  $\pm$  2 % of small granule cells and 30 %  $\pm$  10 % of big granule cells exhibited a specific reddish/brown staining indicative of a high concentration of copper in their granules (**Fig. 5G** [↗](#) and Supp. Fig. S9). These results provide functional proof that SGC are specialized in metal homeostasis in addition to phagocytosis, as suggested by the scRNAseq data (Cluster 3).

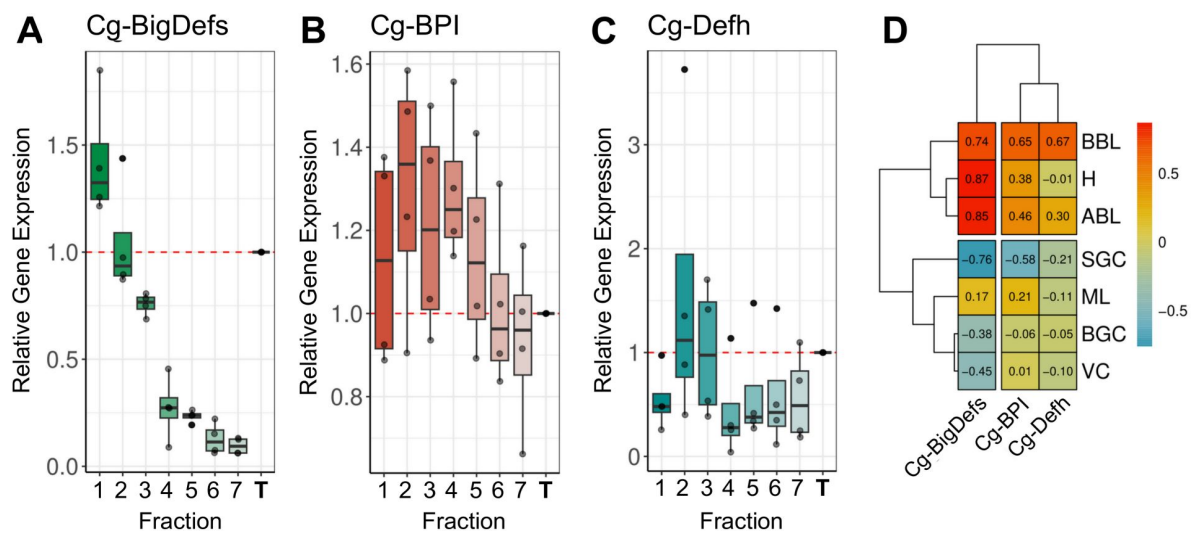
## Antimicrobial peptides are expressed by agranular cells, blasts and hyalinocytes

Antimicrobial peptides (AMPs) have long been studied for their role in the invertebrate humoral immune response. They are expressed by hemocytes, including those in *C. gigas* ([36](#) [↗](#)). The aim was to ascertain whether different hemocyte cell types expressed distinct AMPs.

However, due to the limited sequencing depth, the scRNA-seq analysis was not sensitive enough to reveal AMP expression. This limitation was addressed by investigating AMP expression through RT-qPCR on Percoll density-fractionated hemocytes. The results indicated that Cg-Bigdefs 1-2, Cg-BPI and hemocyte defensin were predominantly expressed by agranular cells, blasts and hyalinocytes (ABL, BBL and H) (**Fig. 6A, B and C** [↗](#)). The expression of these AMPs was associated with Blasts abundance, while the expression of Cg- BigDefs 1-2 was only associated with hyalinocytes (**Fig. 6D** [↗](#)). Granular cells (VC, ML, BGC and SGC) did not seem to express any of the analyzed AMPs. These data suggest that some of the agranular cells are specialized in the production of humoral effectors.

## Tentative model of hemocyte lineages and differentiation pathways in *C. gigas*

The ontogeny, lineage and differentiation pathways of bivalves remain largely unknown ([37](#) [↗](#)). However, there are some indications of circulating and proliferating hemocyte progenitors in the hemolymph of *C. gigas* ([38](#) [↗](#)). GO-terms analysis of the 7 transcriptomic clusters revealed different functional signatures, including the transcriptomic signature of cluster 4, which showed a high expression level of ribosomal proteins (Supp. Fig. S1J). This particularity has been observed in hematopoietic stem cells in vertebrates ([39](#) [↗](#)–[41](#) [↗](#)). Furthermore, scRNA-seq approaches using bioinformatic tools like Monocle3 ([42](#) [↗](#)) can now be used to deduce differentiation pathways from trajectory inference using gene expression profiles analysis. This revealed an



**Fig 6.**

### Hemocyte expression profiles of some antimicrobial peptides.

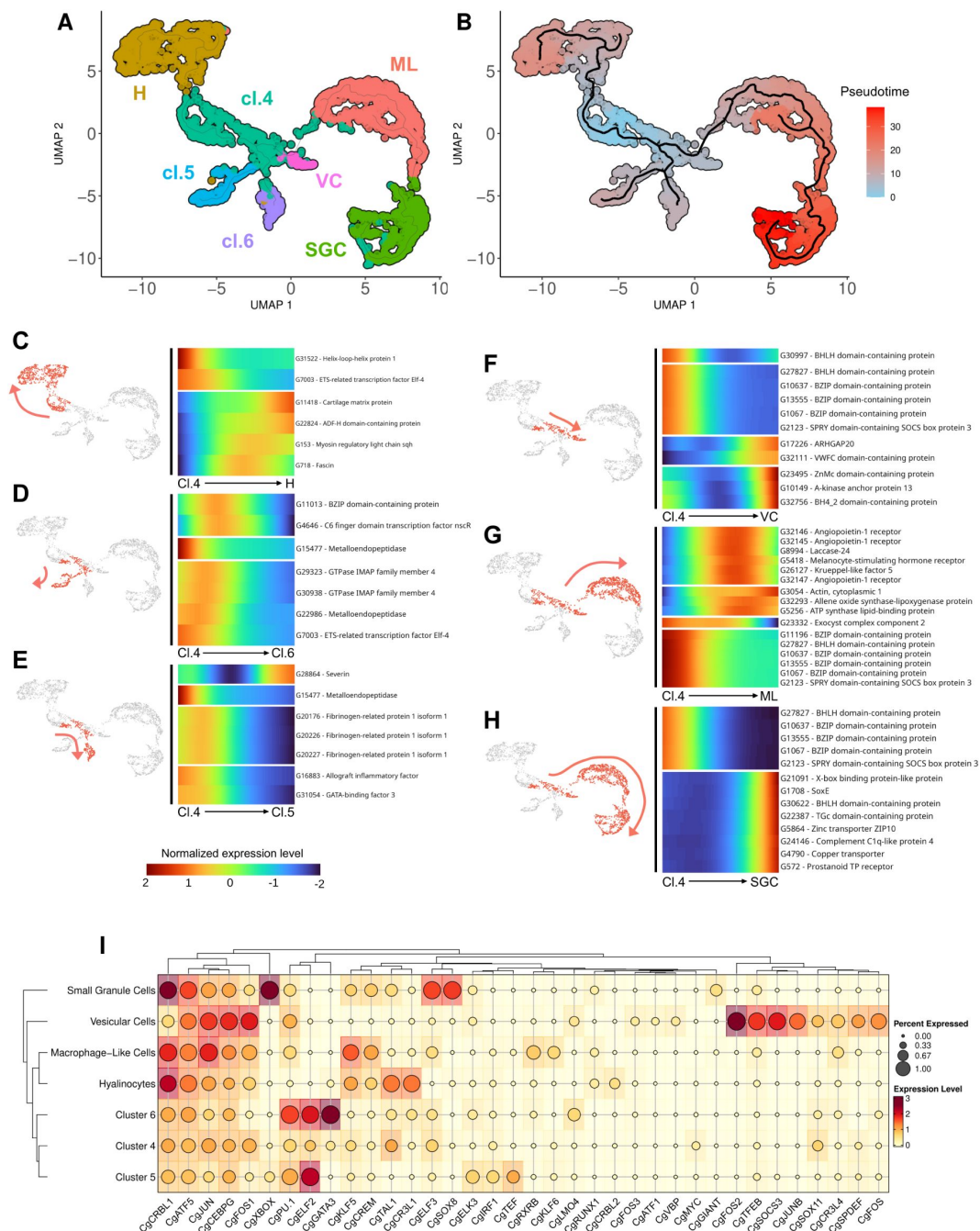
(A), (B) and (C) Relative gene expression in the 7 Percoll hemocyte fractions of Big-Defensin1 & 2 (Cg- BigDefs), BPI (Cg-BPI) and hemocyte defensin (Cg-Defh), respectively, in comparison to the gene expression level in unfractionated hemolymph. Relative gene expression levels were normalized to the reference gene Cg-rps6. The  $2^{-\Delta Ct}$  method was used to calculate relative expression levels, where  $\Delta Ct$  represents the difference between the target gene's Ct value and the reference gene's Ct value. (D) Correlation matrix between the relative gene expression of BigDefensin1 & 2, BPI and hemocyte defensin gene in each fraction and the percentage of each hemocyte type in each fraction (H : Hyalinocytes, ABL : Acidophilic Blast Like, BBL : Basophilic Blast Like, SGC : Small Granule Cell, ML : Macrophage Like, BGC : Big Granule Cell, VC : Vesicular Cell. Values and color scale represent the Pearson correlation coefficient (r) ranging from -1 (inverse correlation) to +1 (full correlation).

overrepresentation of functions involved in the splicing, transcription and translation continuum in the same fourth cluster (see cluster 4 in **Fig. 2A** and **Fig. 2B**), thereby supporting the hypothesis of a pool of quiescent or immature cells that can differentiate upon stimulation.

Cluster 4 was chosen to enroot the pseudotime analysis to deduce differentiation pathways and cell lineages using Monocle3. (**Fig. 7A**). By temporally ordering the 2817 cells analyzed by scRNA-seq, 6 cell lineages could be defined (**Fig. 7B**). Differentiation pathway 1 leads to hyalinocytes (H) (**Fig. 7C**). This transition is characterized by the downregulation of 8 genes (Supp. Fig. S10A), two of which are transcription factors (G31522 and G7003). The hyalinocyte cluster was also characterized by the overexpression of genes involved in cell contractility (G11418, G22824, G153 and G718). Differentiation pathway 2 leads to cells of cluster 5 and is characterized by the downregulation of about 23 genes (Supp. Fig. S10B), including two genes related to metalloendopeptidase activity (G15477 and G22986), two GTPase IMAP genes (G29323 and G30938), and one transcription factor that is specifically downregulated (G7003). Additionally, two other transcription factors are transiently overexpressed (G11013 and G4646) (**Fig. 7D**). Differentiation pathway 3 leads to cells of cluster 6 and is characterized by the down-expression of 21 genes (Supp. Fig. S10C). Three fibrinogen-related protein genes (G20176, G20226 and G20227), a metalloendopeptidase gene (G15477) and a GATA-binding factor (G31054) were downregulated. The *severin* gene was upregulated (G28864) (**Fig. 7E**). The pathways between clusters 4, 5 and 6 found by Monocle3 analysis were pseudo temporally short and few specific markers were identified, suggesting that they were transcriptionally close. Differentiation pathway 4 leads to vesicular cells (VC) (**Fig. 7F**) where a large number of genes were silenced to give rise to these cells (Supp. Fig. S10D). Five genes were specifically overexpressed in VC (G32111, G17226, G32756, G10149 and G23495). Interestingly, 6 genes encoding potential transcription factors were downregulated in this lineage (G30997, G10637, G1067, G13555, G2123 and G27827). Differentiation pathway 5 leads to macrophage-like cells (ML), and is characterized by the underexpression of 35 genes and the overexpression of 13 genes (Supp. Fig. S10E). Among the 35 genes, 6 are putative transcription factors (G2123, G10637, G13555, G27827, G1067 and G11196) (**Fig. 7G**). Finally, we can outline a lineage ending in small granule cells (SGC) (**Fig. 7H**). This pathway involved cluster 4 (immature cells), VC, ML and SGC and was characterized by the downregulation of 27 genes (Supp. Fig. S10F), including 5 potential transcription factors (G10637, G13555, G27827, G1067 and G2123). SGC showed a distinct transcriptomic signature with 61 overexpressed genes, including 3 transcription factors (G1708, G21091 and G30622). Based on these findings, we postulate that immature cluster 4 cells possess pluripotent potential and can give rise to four terminally differentiated cell types : cluster 5 and 6 cells, hyalinocytes and small granule cells, and two other transient hemocyte cell types : vesicular cells and macrophage-like cells. All data from the Monocle3 analyses are available as Supplementary Data (Supp. Fig. S10).

Differentiation pathway analysis thus revealed the over- or under-expression of various transcription factors in the identified pathways. Given the established role of transcription factors as master regulators of cell differentiation and their utility in delineating cell lineages, we investigated the combinatorial expression patterns of transcription factors among the different transcriptomic cell clusters. Based on GO-terms annotation, 28 different sequences corresponding to transcription factors were isolated from the scRNA-seq dataset. The transcription factor function was confirmed by manual annotation (Supp. Table S5) and **Figure 7I** shows the average expression profiles of these factors in the different clusters.

Supplementary Figure S11 illustrates their expression levels in single cells. Two transcription factors, CgATF5 and CgCRBL1, exhibited a contrasting expression profile with an increased average expression in macrophage-like cells, hyalinocytes and small granule cells versus a low expression profile in cells in clusters 4, 5, 6 and vesicular cells. We also identified transcription factors that were specifically expressed in the different transcriptomic clusters : CgSPDEF, CgSOCS3, CgFOS and CgTFEB were specific for vesicular cells, CgSOX8, CgXBOX and CgELF3 were specific for small granule cells, CgCR3L1 and CgTAL1 for hyalinocytes, and CgJUN, CgKLF5, CgKLF6



**Figure 7.**

### Pseudotime ordering of cells revealed 6 potential differentiation pathways of hemocytes.

(A) UMAP plot of scRNA-seq analysis showing the 7 transcriptomic clusters used for pseudotime analysis. 4 clusters were identified cytologically (SGC for small granule cells - cluster 3, H for hyalinocytes - cluster 2, ML for Macrophage Like - cluster 1 and VC for vesicular cells - cluster 7), cl.4, cl.5, and cl.6 represent clusters 4, 5, and 6, respectively. (B) Graphical representation (UMAP projection) of the Monocle 3 pseudo-time order of the clustered cells. Cluster 4 (cl.4) was used as the origin for the pseudotime analysis. (C) (D) (E) (F) (G) and (H) show the gene expression level of selected marker genes obtained from the Monocle3 trajectory analysis at the beginning and end of the modeled differentiation pathways (in red on the UMAP plot) from cluster 4 to hyalinocytes, to cluster 5 cells, to cluster 6 cells, to Vesicular Cells (VC), to Macrophage-Like cells (ML) and to Small Granule Cells (SGC) respectively. The color scale represents the normalized expression level of each gene. (I) Dot plot showing the average expression and the percentage of cells expressing identified transcripts encoding for transcription factors in the scRNA-seq dataset. Expression level expression is expressed in Log2FC and dot diameter represents the percent of cells expressing the gene.



and CgCREM for macrophage-like cells. Eight additional transcription factors were specifically identified in cluster 6 (CgGATA3, CgPU.1 and CgELF2), cluster 5 (CgELF2, CgELK3 and CgIRF1) and cluster 4 (CgTAL1 and CgFOS1). These data potentially define four distinct hematopoietic lineages originating from one type of immature blast cells and give rise to hyalinocytes, SGC (via VC and ML), or two distinct differentiated blast-like cells. We also identified a combination of transcription factors specific to lineages and cell types that are potential master regulators of cell fate during hematopoiesis.

## Discussion

The findings of our study represent a significant advancement in our understanding of the functional diversity and lineages of *C. gigas* hemocytes. Single-cell RNA-seq and cytology were combined to identify seven distinct hemocyte transcriptomic cell clusters and an equivalent number of morphotypes. These include four granular cell types (big granule cells, macrophage-like cells, small granule cells and vesicular cells), two distinct blast-like cells (basophilic and acidophilic blast-like cells), and one agranular epithelial-like cell type (hyalinocytes). A significant challenge was to identify correlations between transcriptomic and cytological data to fully define each cytological cell type. This challenge was overcome by combining multiple approaches, including isopycnic Percoll density gradient fractionation combined with the analysis of transcriptomic markers expression, and functional assays including phagocytosis, oxidative burst, copper accumulation, AMP expression and finally pseudotime analysis of gene expression. These results confirmed the historical classification of the 3 main cell groups : blasts, hyalinocytes, and granular cells (15) and deepened our understanding of the functional specificities of poorly characterized cell types. In particular, we identified distinct transcriptional and functional subtypes among blasts and granular cells with complementary immune specialization and lineage relationships between cells.

One significant outcome of the present study is the identification of cell types involved in antimicrobial activities, including phagocytosis, intracellular copper accumulation, an oxidative burst and antimicrobial peptide production. These cell types have been extensively studied for their role in antibacterial and antiparasitic defenses, as they are found in the large majority of invertebrates (17). Nevertheless, there has been considerable debate surrounding the cell types specialized for these critical immune functions in bivalves, particularly oysters. Moreover, the involvement of the different hemocyte subpopulations in immune functions is not yet fully understood.

Our findings reveal that the macrophage-like (ML) cells and small granule cells (SGC) are the sole hemocyte cell types that function as professional phagocytes, as demonstrated against Zymosan or *Vibrio*. These two distinct cell types could be distinguished functionally. First, two distinct transcriptomic clusters were identified for each cell type (Cluster 1 for ML / BGC and Cluster 3 for SGC). Secondly, only ML induced a measurable oxidative burst. Thirdly, only SGC accumulated intracellular copper in specific granules. Interestingly, our scRNA-seq analysis indicates that SGC (cluster 3) expresses the scavenger receptor cysteine-rich (SRCR) gene G3876, annotated as an Low-density lipoprotein receptor-related protein with a Log2 fold change (Log2FC) of 0.77 linking them to scavenger receptor-mediated pathogen recognition and clearance. This aligns with findings by Wang et al. (43), who demonstrated significant expansion and dynamic regulation of SRCR genes in response to pathogen-associated molecular patterns. The pseudotime trajectory linking macrophage-like (ML) and small granule-like (SGC) cells suggests a potential developmental relationship within the granular cell lineage; however, this hypothesis requires further validation. The notion that ML could serve as a precursor for SGC may seem counterintuitive. However, this is not an isolated phenomenon among invertebrate hemocytes. For instance, in *Drosophila* larvae (44), some populations of professional phagocytes, the sessile plasmatocytes, give rise to crystal cells or lamellocytes that are morphologically and functionally distinct from plasmatocytes (45).

Similarly, the existence of multiple professional phagocytes in the oyster is reminiscent of vertebrate macrophages, polynuclear neutrophils and dendritic cells, which possess distinct functional specializations, including efferocytosis (46), oxidative burst, NETosis, or antigen presentation (47).

The characterization of professional phagocytes in oysters is of particular importance for a deeper understanding of oyster-*Vibrio* interactions during pathogenesis. Some of the most extensively studied oyster pathogens, including strains of *V. tasmaniensis* and *V. aestuarianus francensis*, harbor virulence traits that enable them to disrupt the phagocytic activity of hemocytes during pathogenesis. For instance, *V. tasmaniensis* behaves as a facultative intracellular pathogen with phagocytosis-dependent cytotoxicity (11). Additionally, both *V. tasmaniensis* and *V. aestuarianus* display resistance to copper toxicity through CopA and CusA/B transporters. This trait is essential for the survival and virulence of these pathogens in oysters (9, 48). Since SGCs have been demonstrated to be professional phagocytes with copper-rich granules, the cellular interactions between SGCs and these vibrios are likely to be critical during the antibacterial host response and pathogenesis. ML are phagocytes that possess a very potent NADPH-oxidase-dependent oxidative burst. The oxidative burst is a rapid and potent antimicrobial response observed in professional phagocytes, such as polynuclear neutrophils in mammals. It is worth noting that NADPH-dependent ETosis has been observed in *C. gigas* in a manner analogous to that observed in human neutrophils (49). This cell death is characterized by the projection of DNA extracellular traps that capture and kill some pathogens like *Vibrio* (49). Therefore, it is reasonable to hypothesize that ML may also be involved in ETosis.

The specialized functions of the two other types of granular cells, the BGCs and the VCs, remain unclear. Despite the difficulty in identifying a specific scRNAseq transcriptomic cluster for BGCs, the level of expression of laccase 24 was found to be higher in a particular subcluster among ML (Supp. Fig. S8) and pseudo-time analysis highlighted the same subcluster as an alternative differentiation state among ML (Fig. 7B). The enrichment of transcripts involved in oxidation-reduction pathways, particularly laccase 24, aligns with their potential role in melanization and response to oxidative stress (50). While melanin-like deposits have been observed in some cases of infestation by the parasites *Martelia* or *Bonamia* in the Sydney rock oyster (51), this mechanism is not as robust as that described in arthropods, which perform melanization through a prophenoloxidase activation cascade (52, 53). In many marine invertebrates (50), a type of hemocyte known as Brown Cells could be related to the BGCs described here. When observed without any staining (as in Fig. 5F), their big granules with a yellow to dark brown content appear to align with the historical description of brown cells that often infiltrate tissues (like gills) in animals exposed to polluted waters (50). It has therefore been theorized that these cells are involved in detoxification processes. Our pseudotime analysis indicates that they likely originate from ML, with limited phagocytosis activity and a specialized role in melanization and potentially heavy metal detoxification, as evidenced by rhodanine staining showing copper accumulation in some of their granules (Fig. 5G). Further studies are recommended to clarify the role of BGCs, particularly in the context of parasitic infestation or exposure to toxic stresses. Lastly, VC are granular cells that remain to be functionally characterized. Their transcriptomic profile suggests strong intracellular vesicular trafficking and autophagy activity. However, our functional assays did not reveal any particular immune-related function. Their clear granules appear autofluorescent when illuminated with UV light (Supp. Fig. S12) but the biochemical nature of the content of these granules remains to be characterized. As they also appear as cell intermediates along the granular cell differentiation pathway in pseudotime analysis, they could represent functionally immature precursors of the other three granular cell types, much like promyelocytes which possess specific azurophilic granules but are functionally immature precursors during granulocyte differentiation in humans. However, the enrichment of autophagy-related transcripts in VC calls for further investigation into a potential role in antiviral immunity, as autophagy has been suggested to play a role in the response to OsHV-1 virus (54).

Hyalinocytes are a homogenous cell type, with only one morphotype matching one transcriptomic cluster. It can be deduced from pseudotime analysis that they originate from a specific and very different differentiation pathway than the granular cells. According to the literature, hyalinocytes are involved in the early stages of inflammation and can infiltrate wounds and interact with foreign particles. In the flat oyster *Ostrea edulis*, they contribute to shell production and wound healing (55 [↗](#)). In the Sydney rock oyster they play a role in cell aggregation (56 [↗](#)), while in *C. virginica* (57 [↗](#)) they contribute to encapsulation, reminiscent of lamellocytes in *Drosophila*. Our results suggest an important role of AMP expression in the immune response. Indeed, *Cg-BigDefs*, which participate in the control of oyster microbiota (58 [↗](#)), were found to be expressed in both hyalinocytes and blast-like cells. Moreover, hyalinocytes from the oyster *O. edulis* have been shown to express the AMP Myticin C (59 [↗](#)), which lends further support to this immune function. Among the AMPs, we also found that *Cg-BPI* and *Cg-Defh*, are more expressed in BBL, ABL than in other cell types. These results are somewhat unexpected, given the prevailing assumption that AMPs are stored in granules of granular cells, rather than agranular cells (16 [↗](#)). These results highlight the necessity to reassess the role of specific agranular cell types in the active production of humoral immune effectors. Nevertheless, further staining approaches would be necessary to confirm these transcriptomic results. Our findings suggest that hyalinocytes and/or the blast-like cells may be a cellular target of the OsHV-1 virus, the causal agent of POMS, which dampens the expression of certain AMPs (7 [↗](#)), thereby inducing bacterial dysbiosis. It is still unclear whether this is due to a decreased expression of AMPs and/or inhibition of immature blast cell differentiation involved in the renewal of agranular cell types.

It should be noted that the complexity of blast-like cells could not be fully elucidated in this study, as 3 clusters and only 2 morphotypes were identified. Cell fractionation using Percoll gradient failed to yield pure blast-enriched fractions (ABL and BBL), preventing precise functional characterization. The enrichment in transcripts of the transcription synthesis degradation continuum aligns with the definition of undifferentiated blast-type cells and with the basophilic staining obtained in MCDH for BBL cells, as immature blast cells are characterized by a basophilic cytoplasmic staining in humans. However, our results show that certain blast populations can produce AMPs, suggesting that these cells may also play a role in the production of humoral effectors. Ultimately, it remains unclear whether these circulating immature cells are indeed the stem cells from which all hemocytes originate.

In the animal kingdom, the innate immune system relies on specialized cells derived from pluripotent precursors through hematopoiesis. Transcription factors in particular have been found to exhibit a high degree of conservation throughout the animal kingdom, from invertebrates to vertebrates. However, the mechanisms underlying the functional differentiation of bivalves are only partially available or understood (60 [↗](#)). Recent research has indicated the potential existence of hemocyte progenitors, also known as blast-like cells in several bivalve species. These include clams, mussels, scallops, marine mussels, freshwater mussels, oysters, pearl oysters, and wing shells (60 [↗](#)). However, various models for hematopoiesis in bivalves have been proposed and extensively debated without a clear consensus or definitive proof. Single-cell RNA sequencing and pseudotime analysis have enabled us to propose a refined model of hematopoiesis at an unprecedented level of detail. In this model, the different types of hemocytes are likely produced through four differentiation pathways that originate from a common progenitor. One pathway results in the formation of agranular hyalinocytes, while another independent pathway gives rise to the granular cells, including VC, ML, BGC, and SGC. Furthermore, two additional differentiation pathways have been identified that may lead to terminally differentiated blast-like cells. The differentiation of mature hemocytes involves the establishment of lineage-specific gene expression profiles, which rely on transcription factors to modulate the expression of their target genes. Our study identified a combination of transcription factors that are differentially expressed during *C. gigas* hematopoiesis and are specific to differentiation stages and cell lineages, including many well-known hematopoietic transcription factors such as GATA, PU-1, TAL1 and SOX factors, which are positioned in this model of *C. gigas* hematopoiesis (61 [↗](#)–63 [↗](#)) (Fig. 8 [↗](#)). Nevertheless,

we did not detect any canonical stem or progenitor cell populations in our dataset, underscoring the need for future investigations - potentially involving immunological challenges and lineage-tracing assays - to clarify whether proliferative cells circulate in the hemolymph or instead reside primarily in tissue compartments.

The comparison of hemocyte populations in *C. hongkongensis* and *C. gigas* highlights a shared reliance on granulocytes as primary immune effectors, with *C. hongkongensis* (27 [↗](#)). Our study emphasizes functional specialization in *C. gigas*, focusing on phagocytosis, oxidative stress responses, and vesicular trafficking, likely reflecting differences in transcriptomic and functional characterization methodologies. Moreover, the identification of conserved transcription factors such as Tal1, Sox, Runx and GATA in both datasets underscores the evolutionary conservation of hematopoiesis pathways across metazoans. While direct comparisons between scRNA-seq datasets are hindered by differing annotation frameworks, parallels can still be drawn with studies in other invertebrates. For example, shrimp and crayfish scRNAseq datasets (64 [↗](#)) describe hemocytes with immune-activated states and integrin-related markers, respectively, while *Anopheles gambiae* (65 [↗](#)) exhibits clusters with high ribosomal activity and transcription factors such as bHLH and myc. These findings highlight both the shared molecular foundations and species-specific differences that define hemocyte diversity and specialization.

This study significantly advances our understanding of bivalve immunity, especially in comparison to arthropods. By introducing a standardized reference hemocytogram for oysters using MCDH staining, defining cell type-specific markers and key transcription factors likely involved in cell fate determination, as well as clarifying the functions of different hemocytes, we have paved the way for future in-depth studies. Moreover, the identified clusters represent functionally and transcriptionally distinct cell populations under the conditions studied, yet they may reflect dynamic states rather than fixed subtypes, thereby acknowledging the potential for hemocyte populations to transition between states in response to environmental or physiological changes—even under homeostatic conditions. This will facilitate further studies of the oyster's immune response to various biotic and abiotic stress at the cellular level. Improved comprehension of antiviral and antibacterial responses in bivalves, along with an enhanced understanding of immune priming and immune memory at the cellular level in bivalves, will benefit health and population management practices for sustainable aquaculture production. These findings will also contribute to the broader field of evolutionary immunology by enabling comparative studies and elucidating the diversification of immune cells and immunity-related genes in a protostome.

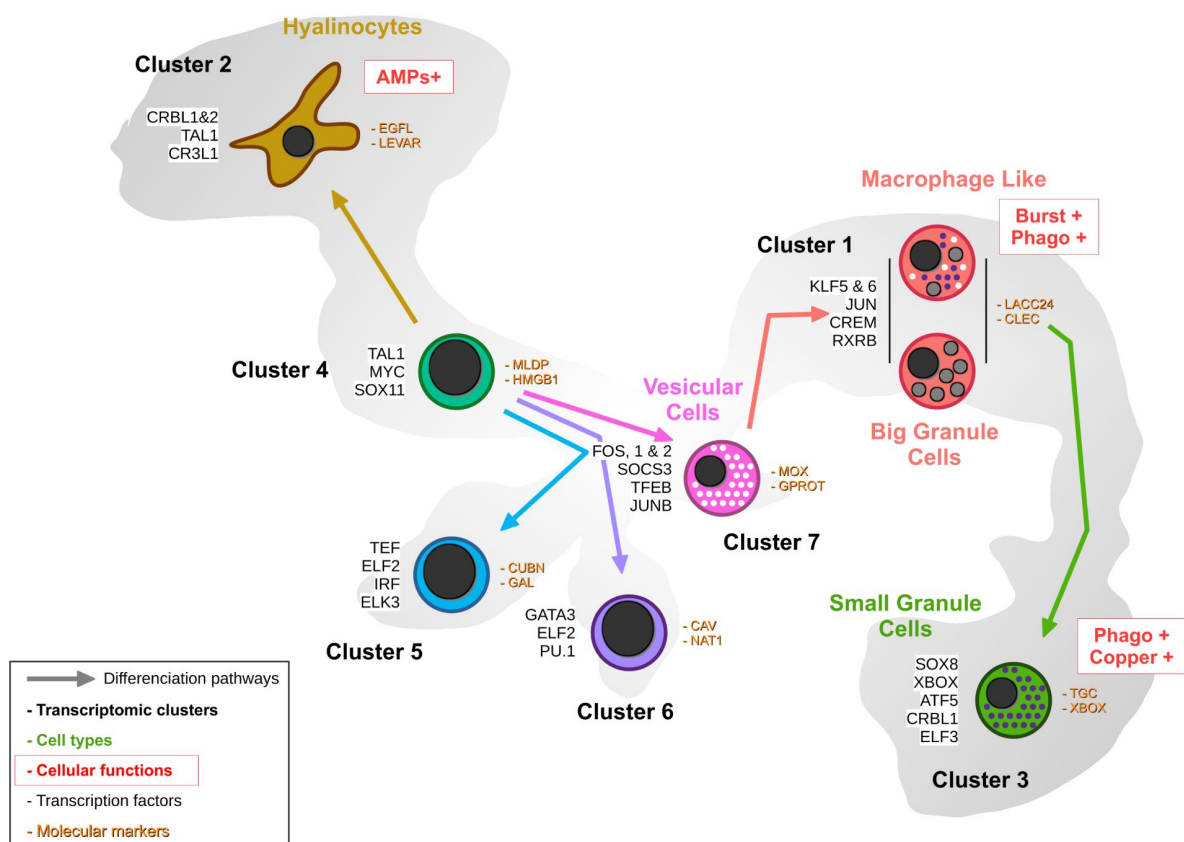
## Materials and Methods

### Conservation of oysters

The work described here was performed using two different sources of oysters of the same species *Crassostrea (Magallana) gigas*. ISA (Ifremer Standardized Oysters - La tremblade - France) oysters for the scRNA-seq experiment and oysters provided by a local supplier (<https://www.huitres-bouzigues.com> [↗](#)). Animals were washed and kept in 10 L tanks containing seawater and a bubbler to oxygenate and homogenize the water. Water was changed daily and oyster health was monitored. All animals used in this study were 18 months of age.

### Hemocyte collection and processing

*Crassostrea gigas* hemocytes were collected from live animals by puncture of the adductor muscle. The oyster shell was incised on the posterior side with forceps. Hemolymph was collected using a 23Gx1" needle mounted on a 5 mL syringe prefilled with 2 mL of ice-cold Alsever modified medium (20.8 g glucose – 8 g trisodium citrate - 22.5 g sodium chloride - 0.4 g BSA - pH=7.5). Samples were centrifuged for 4 minutes at 200 g – 4 °C and the supernatant was removed and



**Figure 8.**

**Proposed hemocyte ontology in *Crassostrea gigas* based on the transcriptomic, cytological and functional results obtained.**

Cells are colored according to the same color code as the transcriptomic clusters. Cluster numbers and cell types are indicated. To the left of the cells are the overexpressed transcription factors and to the right are the identified marker genes in each cluster. Functional characteristics of hyalinocytes, macrophage-like cells and small granule cells are marked in red. (**AMP** : AntiMicrobial Peptide, **Burst** : ROS production, **Phago** : phagocytosis)



replaced with 1 mL of fresh Alsever modified medium. Each hemocyte sample was thoroughly checked for quality, counted under a microscope using KOVA (Kova International, USA) slides, and stored on ice prior to processing. For scRNA-seq analysis, resuspended hemocytes were filtered on 30  $\mu$ m filters, counted, the solution was adjusted to  $1.10^6$  cells per mL and stored on ice prior to 10X genomic library preparation.

### ***C. gigas* genome annotation**

The *C. gigas* genome (Genbank GCA\_902806645.1) (30) was used as a reference. Prior to annotation, the longest CDS sequences were extracted from the gff3 file. Annotation was realized using the ORSON script (<https://gitlab.ifremer.fr/bioinfo/workflows/orson>). ORSON combines cutting-edge tools for annotation processes within a Nextflow pipeline. The ORSON script performs a sequence similarity search with BLAST (66) against the Uniprot-Swissprot and Uniprot-trEMBL databases, and functional prediction with InterProScan (67) and eggNOG (68) orthogroup annotation. Interproscan analysis was performed against Pfam, Prosite, CDD, TIGR, SMART, SuperFamily, PRINTS and Hamap databases. Results were collected and processed using Blast2GO (69) for annotation mapping and validation.

### **Drop Seq-based scRNA-seq library generation**

The 10X Genomics protocol, Single Cell 3' Reagent Kits v2 User Guide from the manufacturer (10X Genomics, USA) was followed to prepare gel in emulsion beads (GEM) containing single cells, hydrogel beads, and reverse transcription reagents, perform barcoded cDNA synthesis, and generate sequencing libraries from pooled cDNAs. The concentration of single-cell suspensions was approximately 1000 cells /  $\mu$ L, as estimated by manual counting, and cells were loaded according to the 10X protocol to capture approximately 3000 cells per reaction. Library construction (after GEM digestion) was performed using 10X reagents according to the manufacturer's instructions. Libraries (paired-end reads 75 bp) were sequenced on an Illumina NovaSeq (Illumina, USA) using two sequencing lanes per sample.

### **scRNA-seq analysis**

Reads were aligned to the *C. gigas* reference genome (Genbank GCA\_902806645.1) (27) using STAR solo software (v 2.7.10) (29). Unique molecular identifiers (UMIs) were extracted and counted for each cell, and an expression matrix was generated for further analysis. Single-cell RNA sequencing (scRNA-seq) data analysis was performed using the R programming language (version 4.2.1) (R Core Team, 2018) and the Seurat package (version 4.3.0) (70). The data were then pre-processed to remove unwanted sources of variation and to normalize gene expression. Cells with small library sizes and a high proportion of mitochondrial genes were excluded. Data normalization was performed using the SCTransform method. After normalization, highly variable genes were identified using the FindVariableFeatures function and the top variable genes were selected for downstream analyses. Dimensionality reduction was performed using Principal Component Analysis (PCA), followed by Uniform Manifold Approximation and Projection (UMAP) to visualize the data. Cell clustering was performed using the 'FindClusters' function, using the previously identified significant principal components (dims = 6) and a resolution parameter ( $r = 0.1$ ) to define cluster granularity.

Differential expression analysis was performed using the 'FindAllMarkers' function ( $pct.min = 0.25$ ) to identify genes differentially expressed between clusters, with statistical significance determined using the Wilcoxon rank sum test. Functional enrichment analysis of differentially expressed genes was performed using gene set enrichment analysis (RGOA) (33).

## KEGG pathway analysis

KEGG analysis was performed using DAVID Bioinformatics Resources (NIAID/NIH) ([32](https://david.ncifcrf.gov/)). Gene lists of specifically overexpressed genes in each cluster were obtained after scRNA-seq processing (genes with  $\text{Log}_2\text{FC} > 0.25$  and significant p-value  $< 0.001$ ) and used for KEGG annotation. The *C. gigas* reference genome from the DAVID bioinformatics resource was used for this analysis, with thresholds of 2 for counts and 0.05 for EASE value (p-value).

KEGG annotation results were post-processed and presented as a heatmap showing the KEGG pathway, fold enrichment, p-value significance and number of positive terms.

## Rank-Based Gene Ontology Analysis

RBGOA first clusters GOs according to their representative genes to identify the most significant GOs, and then ranks the identified biological processes according to their average expression levels (over all representative genes). Finally, biological processes, molecular functions and cellular components significantly enriched in DEGs are identified by a Mann-Whitney rank test with a strict FDR correction. The files generated by the GO-MWU scripts ([https://github.com/z0on/GO\\_MWU](https://github.com/z0on/GO_MWU)) were post-processed to extract the category names and the fraction indicating the number of “good candidates” relative to the total number of genes belonging to that category. The “good candidates” were the genes that exceeded an arbitrary ‘absValue’ cutoff (defined as 0.001) in their significance measure. The results were presented as a heatmap.

## Percoll Density Gradient Separation of Hemocytes10.3390/cells13242106

A concentration series of Percoll (Cytiva, Sweden) diluted in Alsever modified medium was prepared as follows: 10, 20, 30, 40, 50, 60 and 70 % (vol/vol). Discontinuous density gradients (from 10 % Percoll with a density of 1.0647 to 70 % Percoll with  $d=1.1049$ ) were made using 1.5 mL of each concentration and loaded with 1 mL of the hemocyte suspension corresponding to approximately  $2 \cdot 10^7$  cells. Centrifugation was performed (30 min, 800 g, 4 °C) in a swinging bucket on a Beckman Coulter JS-13.1 rotor (Beckman Coulter, USA). Hemocytes concentrated at each density interface were collected separately with a 70 mm long 20Gx2.75” needle mounted on a 1 mL syringe. The hemocytes were then washed from the Percoll by adding 10 mL of ice-cold Alsever modified medium, pelleted by centrifugation (10 min, 200 g, 4°C) and resuspended in Alsever modified medium or filtered seawater.

## Cytological description of the hemocyte populations

200,000 fresh hemocytes were seeded onto a slide using a Cytospin 4 centrifuge (Thermo Scientific, USA). The samples were then stained using the panoptic MCDH (Micro Chromatic Detection for Hematology) (Cellavision, Sweden) staining protocol. This protocol produces purple hues typical of Romanowsky-Giemsa staining results. After staining, the samples were observed using a LEICA DMR (Leica AG, Germany) transmitted light microscope with a 40x magnification objective. Each slide was imaged and the hemocytes were counted and characterized based on their morphology.

## Real Time-quantitative Polymerase Chain Reaction (RT-qPCR)

Total RNA was extracted using the RNeasy kit (Qiagen, the Netherlands) and cDNA was synthesized from 1 µg total RNA using the Superscript IV kit (ThermoFisher Scientific, USA) with oligo(dT) primers. RT-PCR was performed on LightCycler® 480 thermocycler using the SYBR Green 1 Master kit (Roche, Switzerland). Primers were used at 200 nM. Primer sequences are listed in Supplementary Table S3. Expression was normalized to *Cg-rps6* reference gene. The standard cycling conditions were 95°C pre-incubation for 3 minutes followed by 40 cycles of 95°C for 10

seconds, 60°C for 20 seconds, and 72°C for 25 seconds. Each pair of primers was first tested and validated on a total hemocyte RNA preparation to control melting curves and establish calibration lines.

## Oxidative Burst Assay

The production of reactive oxygen species was quantified by luminescence assay. Briefly, hemocytes from hemolymph puncture or Percoll density gradient were washed once with filtered sterile water. 50  $\mu\text{L}$  of hemocytes were plated in triplicate on a 96-well plate ( $3.10^5$  cells/cm<sup>2</sup>). 50  $\mu\text{L}$  of 40 mM luminol (Merck and Co, USA) was added to each well. After 45 minutes of incubation at room temperature, the oxidative burst was induced by adding 100  $\mu\text{L}$  of zymosan (Merck and Co, USA) at a multiplicity of infection (MOI) of 50:1. The plate was immediately placed in a Berthold Centro XS3 LB 960 luminescence microplate reader (Berthold GmbH, Germany) to measure luminescence emission every 2 minutes for 2 hours.

## Phagocytosis assay

400  $\mu\text{L}$  of filtered, sterile water-washed hemocytes were seeded in a 24-well plate at a concentration of  $3.10^5$  cells/cm<sup>2</sup>. After 15 minutes, 50  $\mu\text{L}$  of zymosan was added to the fractions at an MOI of 20:1. For LMG20012<sup>T</sup> *Vibrio*, 50  $\mu\text{L}$  of bacteria was added to a total hemolymph at an MOI of 5:1. After 1 hour of contact at room temperature, the cells were resuspended and 200  $\mu\text{L}$  of the suspension was applied to a microscope slide using a Cytospin centrifuge. The slides were then stained with MCDH and observed under a LEICA DMR transmitted light microscope with a 40x magnification objective (Leica AG, Germany) to count phagocytic cells.

## NitroBlueTetrazolium (NBT) staining

ROS production was measured using NitroBlue Tetrazolium reduction after zymosan stimulation. Briefly, 1 mL ( $1.10^6$  cells) of hemocyte solution was mixed with 50  $\mu\text{L}$  of filtered NBT solution (15 mg/mL in water). Zymosan was added at a 4:1 MOI and the mixture was incubated at room temperature for 10 minutes on a rocking shaker. Then, 50  $\mu\text{L}$  of each sample was plated onto glass coverslips and observed under a transmitted light microscope LEICA DMR with a 40x magnification objective (Leica AG, Germany) to count NBT-positive cells. The positions of positive NBT cells were recorded prior to MCDH staining to identify hemocyte types.

## Rhodanine copper staining of hemocytes

The storage of copper by hemocytes was examined by Rhodanine staining. Briefly,  $1.10^5$  hemocytes were plated on Superfrost slides using a cytospin and circled with a hydrophobic pen to retain the staining solution. The Copper Stain Kit (SkyTek, USA) was used to stain the hemocytes. As described by the kit manufacturer, one drop of Rhodanine solution was added to 9 drops of acetate to form the working solution. Five drops were placed on the cytospin cells and a 5 mL Eppendorf tube with the cap removed was placed over the cells to prevent evaporation of the working solution. The slide and the balanced Eppendorf tube were placed in a beaker of boiling distilled water for 20 minutes. The slide was then washed with 5 drops of acetate and 3 drops of hematoxylin were placed on the slide for 1 minute at room temperature. The slides were then washed a final time with acetate and observed under a LEICA DMR transmitted light microscope with a 40x magnification objective (Leica AG, Germany) to count rhodanine-positive cells. The positions of rhodanine-positive cells were recorded prior to MCDH staining to identify hemocyte types.

## Pseudotemporal ordering of cells with Monocle3

Cells from the *C. gigas* dataset were analyzed using Monocle3 (<https://github.com/cole-trapnell-lab/monocle3>) (42). The Monocle3 analysis was performed on the Seurat object following the aforementioned processing steps. Clustering information (features, genes, partitions, clusters and

UMAP coordinates) was transferred to a CDS object. The cell trajectory was calculated using the *learn\_graph* function. The *choose\_graph\_segments* function was used to select three lineages. The gene expression along pseudotime data was extracted from the result. Then, the data were used to plot genes along pseudotime in three lineages using ggplot2 v3.4.4 R package and the heatmap was generated using the pheatmap v1.0.12 R package.

## Statistical Analysis

To evaluate differences between samples, a statistical analysis was performed using (version 4.2.1) (R Core Team, 2018) and appropriate packages. All data were examined for normality, and statistical tests were selected accordingly. One-way analysis of variance (ANOVA) was used for normally distributed data. Seven different hemolymph samples were used for cytological analysis. Oysters were provided by our local supplier and approximately 300 hemocytes were counted per sample. Seven independent experiments were performed for Percoll density gradient separation. A Tukey test was used to evaluate the statistical difference between the proportions of hemocyte types. The phagocytic capacity of hemocytes was tested in three independent experiments and statistical differences were evaluated using the Tukey test for both the phagocytic capacity between hemocyte types and the number of particles per phagocyte. Finally, oxidative burst capacity was tested 3 times on Percoll-separated hemocytes. The Tukey test was also used to assess statistical differences between conditions. The null hypothesis was rejected at a significance level of  $p = 0.05$ .

## Data and materials availability

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials. Raw reads are available at ENA under the project accession number PRJEB74031.

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## Additional information

### Ethics

This work did not require ethical approval from a human subject or animal welfare committee.

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## Author contributions

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Writing (review and editing): all authors.

## Declaration of AI use

We have not used AI-assisted technologies in creating this article.

## Additional files

***Supplemental Data S2*** [🔗](#)

***Supplemental Data S1*** [🔗](#)

***Supplemental Data S3*** [🔗](#)

***Supplemental Figures*** [🔗](#)



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**Reviewer #1 (Public review):****Summary**

In this manuscript, De La Forest Divonne et al. build a repertoire of hemocytes from adult Pacific oysters combining scRNAseq data with cytologic and biochemical analyses. Three categories of hemocytes were described previously in this species (i.e. blast, hyalinocyte and granulocytes). Based on scRNAseq data, the authors identified 7 hemocyte clusters presenting distinct transcriptional signatures. Using Kegg pathway enrichment and RBGOA, the authors determined the main molecular features of the clusters. In parallel, using cytologic markers, the authors classified 7 populations of hemocytes (i.e. ML, H, BBL, ABL, SGC, BGC, and VC) presenting distinct sizes, nucleus sizes, acidophilic/basophilic, presence of pseudopods, cytoplasm/nucleus ratio and presence of granules. Then, the authors compared the phenotypic features with potential transcriptional signatures seen in the scRNAseq. The hemocytes were separated in a density gradient to enrich for specific subpopulations. The cell composition of each cell fraction was determined using cytologic markers and the cell fractions were analysed by quantitative PCR targeting major cluster markers (two per cluster). With this approach, the authors could assign cluster 7 to VC, cluster 2 to H, and cluster 3 to SGC. The other clusters did not show a clear association with this experimental approach. Using phagocytic assays, ROS, and copper monitoring, the authors showed that ML and SGC are phagocytic, ML produces ROS, and SGC and BGC accumulate copper. Then with the density gradient/qPCR approach, the authors identified the populations expressing anti-microbial peptides (ABL, BBL, and H). At last, the authors used Monocle to predict differentiation trajectories for each subgroup of hemocytes using cluster 4 as the progenitor subpopulation.

The manuscript provides a comprehensive characterisation of the diversity of circulating immune cells found in Pacific oysters.

**Strengths**

The combination of scRNAseq, cytologic markers and gradient based hemocyte sorting offers an integrative view of the immune cell diversity.

Hemocytes represent a very plastic cell population that has key roles in homeostatic and challenged conditions. Grasping the molecular features of these cells at the single-cell level will help understand their biology.

This type of study may help elucidate the diversification of immune cells in comparative studies and evolutionary immunology.

#### Weaknesses

Several figures show inconsistency leading to erroneous conclusions and some conclusions are poorly supported. Moreover, the manuscript remains highly descriptive with limited comparison with the available literature.

<https://doi.org/10.7554/eLife.102622.2.sa3>

#### Reviewer #2 (Public review):

##### Summary:

This work provides a comprehensive understanding of cellular immunity in bivalves. To precisely describe the hemocytes of the oyster *C. gigas*, the authors morphologically characterized seven distinct cell groups, which they then correlated with single-cell RNA sequencing analysis, also resulting in seven transcriptional profiles. They employed multiple strategies to establish relationships between each morphotype and the scRNAseq profile. The authors correlated the presence of marker genes from each cluster identified in scRNAseq with hemolymph fractions enriched for different hemocyte morphotypes. This approach allowed them to correlate three of the seven cell types, namely hyalinocytes (H), small granule cells (SGC), and vesicular cells (VC). A macrophage-like (ML) cell type was correlated through the expression of macrophage-specific genes and its capacity to produce reactive oxygen species. Three other cell types correspond to blast-like cells, including an immature blast cell type from which distinct hematopoietic lineages originate to give rise to H, SGC, VC, and ML cells. Additionally, ML cells and SGCs demonstrated phagocytic properties, with SGCs also involved in metal homeostasis. On the other hand, H cells, non-granular cells, and blast cells expressed antimicrobial peptides. This study thus provides a complete landscape of oyster hemocytes with functional validation linked to immune activities. This resource will be valuable for studying the impact of bacterial or viral infections in oysters.

The main strength of this study lies in its comprehensive and integrative approach, combining single-cell RNA sequencing, cytological analysis, cell fractionation and functional assays to provide a robust characterization of hemocyte populations in *Crassostrea gigas*.

- (1) The innovative use of marker genes, quantifying their expression within specific cell fractions, allows for precise annotation of different cellular clusters, bridging the gap between morphological observations and transcriptional profiles.
- (2) The study provides detailed insights into the immune functions of different hemocyte types, including the identification of professional phagocytes, ROS-producing cells, and cells expressing antimicrobial peptides.
- (3) The identification and analysis of transcription factors specific to different hemocyte types and lineages offer crucial insights into cell fate determination and differentiation processes in oyster immune cells.
- (4) The authors significantly advance the understanding of oyster immune cell diversity by identifying and characterizing seven distinct hemocyte transcriptomic clusters and morphotypes.

These strengths collectively make this study a significant contribution to the field of invertebrate immunology, providing a comprehensive framework for understanding oyster hemocyte diversity and function.

### Conclusion:

The authors largely achieved their primary objective of providing a comprehensive characterization of oyster immune cells. They successfully integrated multiple approaches to identify and describe distinct hemocyte types. The correlation of these cell types with specific immune functions represents a significant advancement in understanding oyster immunity. The authors are aware of the limitations of their study, particularly with regards to the pseudotime analysis, which provides a conceptual framework for understanding lineage relationships but requires further experimental validation to confirm its findings.

This study is likely to have a significant impact on the field of invertebrate immunology, particularly in bivalve research. It provides a new standard for comprehensive immune cell characterization in invertebrates. The identification of specific markers for different hemocyte types will facilitate future research on oyster immunity. The proposed model of hemocyte lineages, while requiring further validation, offers a framework for studying hematopoiesis in bivalves.

<https://doi.org/10.7554/eLife.102622.2.sa2>

### Reviewer #3 (Public review):

The paper addresses pivotal questions concerning the multifaceted functions of oyster hemocytes by integrating single-cell RNA sequencing (scRNA-seq) data with analyses of cell morphology, transcriptional profiles, and immune functions. In addition to investigating granulocyte cells, the study delves into the potential roles of blast and hyalinocyte cells. A key discovery highlighted in this research is the identification of cell types engaged in antimicrobial activities, encompassing processes such as phagocytosis, intracellular copper accumulation, oxidative bursts, and antimicrobial peptide synthesis.

A particularly intriguing aspect of the study lies in the exploration of hemocyte lineages, warranting further investigation, such as employing scRNA-seq on embryos at various developmental stages.

<https://doi.org/10.7554/eLife.102622.2.sa1>

### Author response:

The following is the authors' response to the original reviews.

#### **Reviewer #1 (Public review):**

##### *Summary:*

*In this manuscript, De La Forest Divonne et al. build a repertory of hemocytes from adult Pacific oysters combining scRNAseq data with cytologic and biochemical analyses. Three categories of hemocytes were described previously in this species (i.e. blast, hyalinocyte, and granulocytes). Based on scRNAseq data, the authors identified 7 hemocyte clusters presenting distinct transcriptional signatures. Using Kegg pathway enrichment and RBGOA, the authors determined the main molecular features of the clusters. In parallel, using cytologic markers, the authors classified 7 populations of hemocytes (i.e. ML, H, BBL, ABL, SGC, BGC, and VC) presenting distinct sizes, nucleus sizes, acidophilic/basophilic, presence of pseudopods, cytoplasm/nucleus ratio and presence of granules. Then, the authors compared the phenotypic features with potential transcriptional signatures seen in the scRNAseq. The hemocytes were separated in a*

density gradient to enrich for specific subpopulations. The cell composition of each cell fraction was determined using cytologic markers and the cell fractions were analysed by quantitative PCR targeting major cluster markers (two per cluster). With this approach, the authors could assign cluster 7 to VC, cluster 2 to H, and cluster 3 to SGC. The other clusters did not show a clear association with this experimental approach. Using phagocytic assays, ROS, and copper monitoring, the authors showed that ML and SGC are phagocytic, ML produces ROS, and SGC and BGC accumulate copper. Then with the density gradient/qPCR approach, the authors identified the populations expressing anti-microbial peptides (ABL, BBL, and H). At last, the authors used Monocle to predict differentiation trajectories for each subgroup of hemocytes using cluster 4 as the progenitor subpopulation.

The manuscript provides a comprehensive characterisation of the diversity of circulating immune cells found in Pacific oysters.

#### Strengths:

The combination of the two approaches offers a more integrative view.

Hemocytes represent a very plastic cell population that has key roles in homeostatic and challenged conditions. Grasping the molecular features of these cells at the single-cell level will help understand their biology.

This type of study may help elucidate the diversification of immune cells in comparative studies and evolutionary immunology.

#### Weaknesses:

The study should be more cautious about the conclusions, include further analyses, and inscribe the work in a more general framework.

#### Reviewer #1 (Recommendations for the authors):

The manuscript provides a comprehensive characterisation of the diversity of circulating immune cells found in Pacific oysters.

#### Major comments:

(1) The introduction would benefit from a clear description of what is known about immune cell development and diversity in this model. The bibliography on the three subtypes origins and properties (i.e. blast, hyalinocyte, and granulocytes) should be described in the introduction.

We thank Reviewer #1 for their valuable comments, which have allowed us to further improve our manuscript. We have enriched the introduction with the following addition (line 79 to 82):

“Blast-like cells are considered as undifferentiated hemocyte types (20), hyalinocytes (21) seem to be more involved in wound repair, and granulocytes, more implicated in immune surveillance. The latter are considered as the main immunocompetent hemocyte types (22).”

(2) The authors mentioned a previous scRNAseq dataset produced in another oyster species. They should compare the two datasets to show the robustness of the molecular signatures determined in the present study. In addition, the authors do not mention markers identified in the literature that could be relevant to characterize the clusters (e.g. inflammatory pathway PMID: 29751033, proliferative markers PMID: 36591234/ PMID: 29317231, granulocyte markers PMID: 30633961 ... list not exhaustive). Overall, the comparison of this manuscript dataset and the available literature is too partial

We appreciate the reviewer's suggestion to compare our dataset with previously published scRNAseq data and to integrate markers from the literature. Below, we address these points in detail.

The transcription factors involved in hematopoiesis, such as Tal1, Sox, Runx, and GATA, are highly conserved across metazoans. These markers were identified in our dataset, consistent with findings in other species (1–3), including the previously mentioned scRNA-seq dataset in *C. hongkongensis* (4). However, defining robust and specific markers for distinct hemocyte types remains an ambitious goal that requires validation across diverse biological contexts - work that is beyond the scope of the present study. Additionally, meaningful comparisons between datasets are constrained by differences in annotation frameworks and the absence of a standardized system for defining hemocyte subtypes. These limitations underscore the need for harmonization efforts to facilitate robust cross-study comparisons. Nonetheless, our dataset provides a strong foundation for future comparative analyses once such standardization is achieved.

In response to the reviewer's comment, we have added a paragraph to the discussion (lines 747 - 760) detailing that we identified conserved transcription factor markers in *C. gigas* and *C. hongkongensis*.

(3) The authors sequenced 3000 cells without providing more comprehensive information/rationale on the analysed population. What is the number of hemocytes found in an adult? What proportion of the whole hemocyte population does this analysis represent? Does it include the tissue-interacting hemocytes? Also, what is the rationale for choosing that specific stage?

We thank the reviewer for their insightful questions regarding the analyzed hemocyte population.

Adult 18-month-old *Crassostrea gigas* contain approximately 1 million circulating hemocytes per mL of hemolymph, with an average of 1 mL of hemolymph per individual. Thus, this represents approximately 1 million circulating hemocytes per oyster. For our scRNA-seq analysis, we sampled 3,000 hemocytes, which corresponds to 0.3% of the total circulating hemocyte population.

The number of cells processed was optimized to minimize the occurrence of doublets during scRNAseq. Following 10x Genomics Chromium guidelines, we loaded 4,950 cells to successfully recover a target of 3,000 cells, with a doublet rate of 2.4%, well below the target threshold of 2.5%. This information has been added on line 125 of the document. The target was 3,000 cells, and as reported in Supplementary Table S1, the estimated number of cells after STAR-solo alignment was 2,937. This ensures the reliability and accuracy of single-cell transcriptomic data.

We selected 18-month-old oysters for two key reasons: (i) to facilitate hemolymph collection, as hemocyte counts are more stable and sufficient at this stage, enabling us to collect enough cells for all planned experiments, including functional and cytological analyses; and (ii) to use oysters that are not susceptible to OsHV-1  $\mu$ Var herpesvirus, which predominantly affects younger animals. This ensured that the hemocyte populations analyzed were not influenced by viral infections or related immune responses.

Our study focused on circulating hemocytes collected from hemolymph, which does not include tissue-interacting hemocytes. While these cells may represent an additional population of interest, they fall outside the scope of our current investigation.

By carefully selecting the animal stage and optimizing cell sampling, we ensured that the scRNA-seq dataset provides a robust representation of circulating hemocyte diversity while



maintaining high data quality.

*(4) For the GO term enrichment analysis, the authors included all genes presenting a cluster enrichment above  $L2FC > 0.25$ . This seems extremely low to find distinct functions for each cluster. The risk is to call "cluster specific GO term" GO terms for which the genes are poorly enriched in the cluster. For the most important GO term mentioned in the text, the authors should show the expression levels of the genes (with DotPlot similar to Fig1D) to illustrate the specificity of the GO term. At last, the GO enrichment scores were apparently calculated using the whole genome as background. The analysis, aiming at finding differences between hemocyte subgroups, should use the genes detected in the dataset as background.*

We appreciate the reviewer's concerns regarding the threshold used for GO term enrichment analysis and the choice of background genes. Below, we provide clarification on these points.

For nuanced comparisons, such as those between activation states of the same cell type, lower thresholds for  $\log_2FC$  (e.g.,  $\geq 0.25$ ) are commonly used to detect subtle regulatory shifts. In single-cell RNA sequencing (scRNA-seq) analyses, it is typical to use a  $\log_2FC$  threshold between 0.25 and 0.5 to ensure that biologically relevant, yet subtle, changes are captured. For our analysis, this threshold was chosen to maintain sensitivity to such shifts, particularly given the diversity and functional specialization of hemocyte clusters.

To address the reviewer's suggestion, we will include DotPlot representations (similar to Fig. 1D) for the most significant GO terms highlighted in the text. This will illustrate the expression levels of the associated genes across clusters and demonstrate their specificity to the identified GO terms.

Regarding the background used in the GO enrichment analysis, we employed the Rank Based Gene Ontology Analysis (RBGOA) approach, which explicitly states in its documentation: "It is important to have the latter two tables representing the whole genome (or transcriptome) — at least the portion that was measured — rather than some select group of genes since the test relies on comparing the behavior of individual GO categories to the whole." Our analysis was conducted in agreement with these initial recommendations, ensuring that the results are consistent with the methodology outlined for RBGOA.

*(5) The authors reannotated the genes of *C. gigas* to reach 73.1% annotation. What are the levels of annotations found prior to the reannotation? What do the scores/scale bars from the RBGOA analysis mean in Figures 2B-D?*

Thank you for your comment. The original annotation for *C. gigas* was based on the work of Penalzo et al. (5), which provided GO annotations for 18,750 out of 30,724 genes, corresponding to 61% annotation. Following our reannotation efforts, we were able to increase the annotation coverage to 73.1%, enhancing the resolution of downstream analyses. In response to the reviewer's comment, we have updated the results section (line 211 and 216) to explicitly include the original annotation coverage of 61% from the work of Penalzo et al., followed by details on our newly achieved annotation percentage of 73.1%.

Thank you for pointing this out. We apologize for the oversight regarding the scale bar in Figures 2BD. The colors in the original figure correspond to a z-score calculated from the gene ratio, which was not clearly explained and may have caused confusion. In the revised version of the manuscript, we propose a new representation to facilitate understanding and improve the clarity of the data presentation (Figure 2B).

*(6) The authors describe first the result of the Kegg enrichment analysis and then of the RBGOA. To gain fluidity, I would suggest merging the results of both Kegg and RBGOA for*

each cluster.

Thank you for the suggestion. To enhance the fluidity of the results section, we have redesigned the KEGG/RBGOA figure (see figure 2A and 2B) to present the results for each cluster in an integrated manner. This revised approach aims to provide a clearer and more cohesive representation of the findings.

*(7) The authors make correlations between gradient fraction containing multiple hemocyte populations and qPCR expression levels of cluster-specific markers to associated cytologic features with specific clusters. If feasible, I would recommend validating the association of several markers with hemocyte subgroups using in situ hybridisation or immunolabelling.*

Cytological identification of hemocytes in our study relies on MCDH staining, which provides detailed morphological and cytological information. Unfortunately, the fixation methods required for in situ hybridization (ISH) or immunolabeling are not compatible with those used for MCDH staining. We attempted to combine these approaches but found that the fixation protocols necessary for ISH or immunolabeling compromised the quality of the cytological features observed with MCDH staining. Consequently, such validation was not feasible within the constraints of our experimental setup.

*(8) Anti-microbial peptides are mentioned as enriched in agranular cells based on the gradient/qPCR analysis (Figure 6). Are these AMPs regulated by inflammatory pathways? Are any inflammatory pathways enriched in any scRNAseq cluster? In addition, without validating the data by directly labelling AMP in the different populations, it seems hard to conclude that AMP are expressed only by agranular cells.*

In oysters, two families of antimicrobial peptides/proteins appear to be transcriptionally regulated in hemocytes in response to an infection. The first is that of Cg-BigDefs (6). A 2020 article indicates that the expression of CgBigDef1 is regulated by CgRel, an ortholog of the NFkB transcription factor, which also control the expression of the proinflammatory cytokine CgIL17 (7). Cg-BPI is induced in response to infection but its regulatory pathways remain unknown (8). The last well characterized family of antimicrobial peptides is Cg-Defs. It exhibits constitutive expression in hemocytes.

In our scRNA-seq analysis, CgRel (G12420) shows an increased expression in cluster 5, with a log2FC of 0.4 (equivalent to a 1.32-fold change or 32% higher expression compared to other clusters). Cluster 5 corresponds to blast-like cells, which are transcriptionally distinct and predominantly found in fractions 1, 2, and 3. These same fractions exhibit the highest CgBigDef expression, as demonstrated by qPCR.

From our qPCR results, we see no expression of the three AMP families in cell-sorted granular cells while the cell-sorted agranular cells are positive for the three AMP families, even for inducible ones. Still, we agree that labelling of cell sorted hemocyte populations would reinforce our data. We now specify in the text that further staining would be necessary to confirm these transcriptomic results (Discussion, lines 695 to 296).

*(9) The authors should play down some statements concerning cluster identity. In the absence of a true lineage tracing approach, it is possible that those clusters represent states rather than true cell subtypes. Immune cells are very plastic in nature and able to adapt to the environment, even in conditions that are considered homeostatic.*

We appreciate the reviewer's insightful comment regarding the plasticity of immune cells and the potential for clusters to represent states rather than distinct cell subtypes. We agree that, in the absence of a lineage tracing approach, definitive classification of clusters as fixed

subtypes is challenging. Immune cells, including those in invertebrates, are known for their high degree of plasticity and adaptability to environmental cues.

In response to the reviewer's comment, we have revised the Discussion section to include a statement clarifying that these clusters may represent dynamic states rather than fixed subtypes, thereby acknowledging the plasticity of immune cells (lines 766 to 770).

*(10) Related to the above issue, there is no indication of stem cells being present in the cell population. Is there any possibility to look for proliferative or progenitor markers? In homeostatic and in challenged conditions (for example Zymosan treatment)? This would provide some hints into the cellular pathways involved in the response. Perhaps determining the number/fraction of phagocytic cells in challenged conditions would help as well, in the absence of time-lapse assays.*

Thank you for highlighting the possibility of stem cells or progenitor markers in our hemocyte populations. In our current analysis, we did not detect any known stem cell or proliferative markers, nor evidence of a clearly defined hematopoiesis site in the hemolymph. Indeed, previous work suggests that oyster hematopoiesis may occur in tissues such as the gills, implying that stem or progenitor cells might not circulate in the hemolymph under homeostatic conditions. Consequently, it is plausible that our observation of no proliferative cell populations partly reflects their absence in hemolymph, especially in naïve (unstimulated) oysters. To conclusively identify potential progenitor cells and their proliferative activity, further approaches involving deliberate perturbation of hemocyte homeostasis - such as immunological challenge (e.g., Zymosan treatment) combined with lineage tracing or proliferation assays - would be necessary. These future investigations would not only clarify whether proliferative cells emerge in the hemolymph in response to environmental or pathological stimuli but also help elucidate the broader cellular pathways underlying oyster immune responses.

In response to the reviewer's comment, we have revised the Discussion (lines 742 to 745) and added : "Nevertheless, we did not detect any canonical stem or progenitor cell populations in our dataset, underscoring the need for future investigations - potentially involving immunological challenges and lineage-tracing assays - to clarify whether proliferative cells circulate in the hemolymph or instead reside primarily in tissue compartments."

*(11) Could the authors discuss the phagocytic hemocytes in light of scavenger receptor expression?*

We thank the reviewer for this insightful question. Our study identifies macrophage-like cells and small granule cells as the principal phagocytes in *Crassostrea gigas*, capable of robust pathogen engulfment. Transcriptomic data reveal that these cell types express markers associated with endocytosis and immune defense pathways, such as CLEC and LACC24, which are integral to their phagocytic functionality.

Interestingly, our single-cell RNA sequencing analysis indicates that cluster 3, corresponding to small granule cells, expresses the scavenger receptor cysteine-rich (SRCR) gene G3876, annotated as an Low-density lipoprotein receptor-related protein with a Log2 fold change (Log2FC) of 0.77. This finding directly links small granule cells to scavenger receptor-mediated functions, supporting their role as professional phagocytes. Scavenger receptors, including SRCR proteins, are known for their ability to bind and internalize diverse ligands, including pathogens, and their presence in small granule cells highlights a potential mechanism for pathogen recognition and clearance.

Additionally, scavenger receptors are significantly expanded in oysters, as shown in Wang et al. (9). These receptors exhibit dynamic upregulation in hemocytes upon pathogen exposure,

particularly following stimulation with pathogen-associated molecular patterns (PAMPs) such as lipopolysaccharide (LPS). This evidence suggests that SRCR proteins, including the one identified in our study, play a pivotal role in the phagocytic activities of hemocytes by facilitating pathogen recognition and internalization.

We propose to add this paragraph (lines 610 to 618) in the Discussion : “Interestingly, our scRNA-seq analysis indicates that SGC (cluster 3) expresses the scavenger receptor cysteine-rich (SRCR) gene G3876, annotated as an Low-density lipoprotein receptor-related protein with a Log2 fold change (Log2FC) of 0.77 linking them to scavenger receptor-mediated pathogen recognition and clearance. This aligns with findings by Wang et al. (9), who demonstrated significant expansion and dynamic regulation of SRCR genes in response to pathogen-associated molecular patterns. “

*(12) I am not convinced by the added value of the lineage analysis and the manuscript could stand without it. There is no experimental validation to substantiate the filiation between the clusters. In addition, rooting the lineage to cluster 4 is poorly justified (enrichment in the ribosomal transcript). Cluster 6 is also enriched in ribosomal transcripts and this enrichment can be caused by the low threshold used for the selection of cluster-specific genes (L2FC >0.25). At last, cluster 4 > VC and cluster 4 >SGC belong to the same lineage according to Figure 7 FH.*

We thank the reviewer for their detailed comments regarding the lineage analysis. We acknowledge the limitations in experimentally validating the proposed filiation between clusters, as hemocytes in *Crassostrea gigas* cannot currently be cultivated *ex-vivo*, and we lack the ability to isolate cells specifically from cluster 4 for further functional assays. Consequently, our lineage analysis is based solely on transcriptomic data and pseudo-time trajectory analysis.

Hematopoietic stem cells (HSCs) are a population of stem cells that are largely cell-cycle-quiescent (G0 phase) with low biosynthetic activity. Upon stimulation and stress HSCs undergo proliferation and differentiation and produce all lineages of hemocytes.

Ribosomal proteins play a multifaceted role in preserving the balance between stem cell quiescence and activation. By ensuring precise regulation of protein synthesis, they allow stem cells to maintain their undifferentiated state while remaining poised for activation when needed. Furthermore, ribosomal proteins contribute to the cellular stress response, safeguarding stem cells from oxidative damage and other stressors that could compromise their functionality. Importantly, ribosomal biogenesis and the dynamic assembly of ribosomes provide a regulatory mechanism that fine-tunes the transition from self-renewal to differentiation, a critical feature of hematopoietic stem cells (HSCs) and other stem cell types. These mechanisms collectively highlight the indispensable role of ribosomal proteins in stem cell biology, underscoring their relevance to our study's findings.

In vertebrate, the maintenance of hematopoietic stem cells (HSCs) and hematopoietic homeostasis is widely acknowledged to rely on the proper regulation of ribosome function and protein synthesis (10). This process necessitates the coordinated expression of numerous genes, including genes that encode ribosomal proteins (RP genes) and those involved in regulating ribosome biogenesis and protein translation. Disruptions or mutations in these critical genes are associated with the development of congenital disorders (11). Among these, Rpl22 (found in cluster 4 with a Log2FC of 1.59) has been shown to play a pivotal role in HSC maintenance by balancing ribosomal protein paralog activity, which is critical for the emergence and function of HSCs (12).

Regarding the justification for rooting the lineage to cluster 4, our decision was informed by the enrichment of ribosomal transcripts and functional annotations suggesting a role in translation and cell proliferation, consistent with a precursor-like state. The use of a log2 fold-

change (L2FC) threshold of  $>0.25$ , while conservative, allowed us to include subtle but meaningful transcriptional shifts essential for resolving lineage transitions.

Finally, the lineage progression from cluster 4 to vesicular cells (VC), macrophage-like cells (ML), and ultimately small granule cells (SGC) is supported by trajectory analysis (Figure 7FH), which consistently places VC and ML as intermediates in the differentiation process toward SGC. Although experimental validation is currently not feasible, these findings provide a conceptual framework for future investigations when cell isolation and functional validation tools become available.

*(13) The figures containing heatmaps (Figure 7, Figure 2, Figure S10) or too many subpanels (Figure S5) and Table S5 are hardly readable.*

Thank you for highlighting the issues related to the clarity of the heatmaps (Figures 2, 7, and S10), the multi-panel figure (Figure S5), and Table S5. In response to your feedback, we have revised all of these elements to enhance readability and comprehension. Specifically, we increased font sizes, optimized color scales, and reorganized the layout of the subpanels to emphasize the key findings. We also updated Table S5 to ensure that the data are presented in a clear and easily interpretable format.

We trust that these modifications address the concerns raised and improve the overall clarity of the figures and table.

*(14) A number of single-cell analyses are now available in different species and the authors allude to similar pathways/transcription factors being involved. Perhaps the authors could expand on this in the discussion section.*

Transcription factors involved in hematopoiesis, such as Tal1, Runx and GATA, are highly conserved across metazoans. Consistent with findings in other species, our dataset identifies these markers, reinforcing the evolutionary conservation of these pathways. Furthermore, these markers are also reported in the previous scRNA-seq dataset for *C. hongkongensis* (4), supporting the robustness of our molecular signatures. However, defining specific and robust markers for distinct hemocyte types remains an ambitious task, requiring additional validation in diverse biological and experimental contexts. This validation is beyond the scope of the present study.

In addition, meaningful comparisons between scRNA-seq datasets are constrained by differences in annotation frameworks and the absence of standardized definitions for hemocyte subtypes. Harmonizing these datasets to enable robust cross-species comparisons is a critical challenge for future studies. Nonetheless, the insights provided by our dataset establish a strong foundation for such comparative analyses when these standardization efforts are realized.

In crayfish (1), 16 transcriptomic clusters were identified corresponding to three hemocyte types, with markers such as integrin prominently expressed in hyalinocytes, consistent with our identification of integrin-related genes in hemocytes. In shrimp (1), 11 transcriptomic clusters were described, with markers of hemocytes in immune-activated states, that we observed also in our dataset. For *Anopheles gambiae* (2), 8 transcriptomic clusters were identified, including clusters with high ribosomal activity, analogous to those we described in our study. Finally, in *Bombyx mori* (3), 20 transcriptomic clusters were reported, corresponding to five cytological hemocyte types. Transcription factors such as bHLH, myc, and runt were identified in granulocytes and oenocytoid, showing parallels with markers identified in our dataset.

Despite these similarities, cross-species comparisons are hindered by variability in genome availability and annotation quality, which complicates the precise identification and

functional characterization of genes across datasets. Notably, we did not detect prophenoloxidase genes in our dataset, unlike shrimp and crayfish, suggesting potential species-specific differences in immune mechanisms.

Regarding the previously published *C. hongkongensis* scRNA-seq dataset (4), we observe overlap in markers such as runx and GATA. However, direct comparisons remain limited due to differences in dataset annotations and definitions of hemocyte subtypes. This underscores the need for standardized frameworks to facilitate cross-study comparisons. While we emphasize that robust cross-species validation was beyond the scope of this study, our findings contribute valuable insights into the molecular signatures of oyster hemocytes and provide a framework for future comparative research.

We have expanded our discussion to include comparisons with available scRNAseq data from other invertebrate species (lines 747 to 760)

*Minor comments:*

*(1) Figure 2A-D: to increase the readability of the figure, the authors should display only the GO terms mentioned in the text and keep the full list in supplementary data.*

To enhance the fluidity of the results section, we have redesigned the KEGG/RBGOA figure to present the results for each cluster in an integrated manner (See figure 2A and 2B).

*(2) Line 223: the authors mention that cluster 1 is characterized by its morphology without providing an explanation or evidence.*

We have revised the description of Cluster 1 to remove references to morphology, ensuring consistency with the data presented at this stage of the manuscript (lines 227 to 229): "Cluster 1, comprising 27.6 % of cells, is characterized by GO-terms related to myosin complex, lamellipodium, membrane and actin cytoskeleton remodelling, as well as phosphotransferase activity."

*(3) Line 306: the authors mentioned expression levels and associated them with Log2FC, which represents an enrichment, not the level of expression.*

Thank you for pointing this out. We agree that log2FC represents enrichment rather than absolute expression levels. We have revised the text in the manuscript to clarify this distinction (line 309). The corrected text now states that log2FC reflects the degree of enrichment or depletion of a gene in a specific cluster relative to others, rather than its absolute expression level.

*(4) Figure 4B: the figure shows the distribution of all hemocytes subgroups for each fraction. To better appreciate the distribution of the subgroups in the different fractions, it would be good to have the number of cells of each subtype in the fractions.*

We thank the reviewer for their suggestion to include the number of cells of each subtype in the fractions. While we do not have the exact total number of cells per fraction, we systematically performed hemocyte counts for each fraction as part of our methodology. These counts provide a robust estimation of hemocyte distributions across fractions.

Including these counts in the figure could be an alternative approach; however, we believe it would not significantly enhance the interpretability of the data, as the focus of this analysis is on the relative proportions of hemocyte subtypes rather than absolute numbers. The current representation provides a clear and concise overview of subtype distribution patterns, which aligns with the goals of the study.



Nevertheless, if the reviewer considers it essential, we are open to integrating the hemocyte counts into the figure or supplementing the information in the text or supplementary materials to provide additional context.

*(5) Line 487-488: the authors mentioned that monocle 3 can deduce the differentiation pathway from the mRNA splice variant. I did not find this information in the publication associated with the statement.*

Thank you for pointing this out. We acknowledge the inaccuracy in our statement regarding Monocle3's capabilities. Monocle3 does not deduce differentiation pathways based on mRNA splice variants, as was erroneously suggested in the manuscript. Instead, Monocle3 performs trajectory inference using gene expression profiles. It calculates distances between cells based on their transcriptomic profiles, where cells with similar profiles are positioned closer together, and those with distinct profiles are farther apart. This method enables the construction of potential differentiation trajectories by identifying paths between transcriptionally related cells.

We revise the text in the manuscript to accurately describe this process and remove the incorrect reference to mRNA splice variants (lines 495 to 497).

*(6) Figures 6C-H display heatmaps with two columns representing the beginning and the end of the lineage predicted. It would be more talkative to show the whole path presented in Figure S10.*

Thank you for pointing out that Figures 7C–H currently only show the beginning and end of the predicted lineage, limiting the clarity of the intermediate stages. In response to your suggestion, we have revised these figures to include the full trajectory as presented in Figure S10, ensuring that the intermediate transitions are more clearly visualized. We believe these modifications offer a more comprehensive overview of the entire lineage and enhance the interpretability of our results.

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#### **Reviewer #2 (Public review):**

##### *Summary:*

*This work provides a comprehensive understanding of cellular immunity in bivalves. To precisely describe the hemocytes of the oyster *C. gigas*, the authors morphologically characterized seven distinct cell groups, which they then correlated with single-cell RNA sequencing analysis, also resulting in seven transcriptional profiles. They employed multiple strategies to establish relationships between each morphotype and the scRNAseq profile. The authors correlated the presence of marker genes from each cluster identified in scRNAseq with hemolymph fractions enriched for different hemocyte morphotypes. This approach allowed them to correlate three of the seven cell types, namely hyalinocytes (H), small granule cells (SGC), and vesicular cells (VC). A macrophage-like (ML) cell type was correlated through the expression of macrophage-specific genes and its capacity to produce reactive oxygen species. Three other cell types correspond to blast-like cells, including an immature blast cell type from which distinct hematopoietic lineages originate to give rise to H, SGC, VC, and ML cells. Additionally, ML cells and SGCs demonstrated phagocytic properties, with SGCs also involved in metal homeostasis. On the other hand, H cells, nongranular cells, and blast cells expressed antimicrobial peptides. This study thus provides a complete landscape of oyster hemocytes with functional validation linked to immune activities. This resource will be valuable for studying the impact of bacterial or viral infections in oysters.*

##### *Strengths:*

*The main strength of this study lies in its comprehensive and integrative approach, combining single-cell RNA sequencing, cytological analysis, cell fractionation, and functional assays to provide a robust characterization of hemocyte populations in *Crassostrea gigas*.*

*(1) The innovative use of marker genes, quantifying their expression within specific cell fractions, allows for precise annotation of different cellular clusters, bridging the gap between morphological observations and transcriptional profiles.*

(2) The study provides detailed insights into the immune functions of different hemocyte types, including the identification of professional phagocytes, ROS-producing cells, and cells expressing antimicrobial peptides.

(3) The identification and analysis of transcription factors specific to different hemocyte types and lineages offer crucial insights into cell fate determination and differentiation processes in oyster immune cells.

(4) The authors significantly advance the understanding of oyster immune cell diversity by identifying and characterizing seven distinct hemocyte transcriptomic clusters and morphotypes.

These strengths collectively make this study a significant contribution to the field of invertebrate immunology, providing a comprehensive framework for understanding oyster hemocyte diversity and function.

#### Weaknesses:

(1) The authors performed scRNAseq/lineage analysis and cytological analysis on oysters from two different sources. The methodology of the study raises concerns about the consistency of the sample and the variability of the results. The specific post-processing of hemocytes for scRNAseq, such as cell filtering, might also affect cell populations or gene expression profiles. It's unclear if the seven hemocyte types and their proportions were consistent across both samples. This inconsistency may affect the correlation between morphological and transcriptomic data.

We thank the reviewer for highlighting the importance of sample consistency and potential variability, and we acknowledge the need for clarification regarding the use of oysters from two different sources.

Oysters from La Tremblade (known pathogen-free in standardized conditions) were used to establish the hemocyte transcriptomic atlas through scRNA-seq and for cytological analyses. Oysters from the Thau Lagoon (Bouzigues) were used for cytological, functional, and fractionation experiments. These oysters were sampled during non-epidemic periods and monitored under Ifremer's microbiological surveillance to ensure pathogen free status.

The cytological results (hemocytograms) presented in Figure 3 and Supplementary Figure S3 were derived from Thau Lagoon oysters. To clarify, we updated The Table 3 in Figure 3 and Supplementary Figure S3 to explicitly display hemocyte counts for oysters from both La Tremblade and Thau Lagoon. These data confirm consistent proportions of hemocyte types across both sources, with no significant differences ( $p > 0.05$ ).

Hemocyte isolation and filtering protocols were rigorously optimized to preserve cell viability and morphology during scRNA-seq library preparation. Viability assays and cytological evaluations confirmed that these procedures did not significantly alter hemocyte populations or their proportions. Sample processing times were minimized to ensure that the scRNA-seq results accurately reflect the native state of the hemolymph.

Taken together, our results confirm that variability between oyster sources or methodological processes did not compromise our findings. This ensures that the correlations between morphological and transcriptomic data are reliable and robust.

(2) The authors claim to use pathogen-free adult oysters (lines 95 and 119), but no supporting data is provided. It's unclear if the oysters were tested for bacterial and viral contaminations, particularly *Vibrio* and OshV-1  $\mu$ Var herpesvirus.

The oysters used in this study were sourced from two distinct origins. First, the animals (18 months old) utilized for scRNA-seq and cytological analyses were obtained from the Ifremer controlled farm located in La Tremblade, France (GPS coordinates: 45.7981624714465, -1.150171788447683). This facility exclusively produces standardized oysters bred in controlled conditions with filtered seawater, entirely isolated from environmental known pathogens. The oysters from this source are certified “pathogen-free” upon arrival at the laboratory, following Ifremer’s stringent quality control protocols. We have replaced the term ‘pathogen-free’ with ‘known pathogen-free’ (line 123) to accurately reflect the animals’ true status.

Second, for the fractionation experiments and functional tests, oysters were either sourced from the aforementioned Ifremer farm or from a producer located in the Thau Lagoon, France (GPS coordinates: 43.44265228308842, 3.6359883059292057). The Thau Lagoon is subject to comprehensive environmental and microbiological surveillance by the Ifremer monitoring network and the regional veterinary laboratory. For these experiments, we specifically selected oysters aged 18 months - an age associated with reduced susceptibility to OsHV-1  $\mu$ Var herpesvirus - and ensured that sampling occurred outside of any detected epidemic periods. Furthermore, prior to experimentation, hemocyte samples from all oysters were examined. Oysters showing signs of contamination or exhibiting abnormal hemocyte profiles were excluded from the study.

These measures ensured that the oysters used in this work were of high health status and minimized the likelihood of bacterial or viral contamination, including *Vibrio* and OsHV-1  $\mu$ Var.

*(3) The KEGG and Gene Ontology analyses, while informative, are very descriptive and lack interpretation. The use of heatmaps with dendrograms for grouping cell clusters and GO terms is not discussed in the results, missing an opportunity to explore cell-type relationships. The changing order of cell clusters across panels B, C, and D in Figure 2 makes it challenging to correlate with panel A and to compare across different GO term categories. The dendrograms suggest proximity between certain clusters (e.g., 4 and 1) across different GO term types, implying similarity in cell processes, but this is not discussed. Grouping GO terms as in Figure 2A, rather than by dendrogram, might provide a clearer visualization of main pathways. Lastly, a more integrated discussion linking GO term and KEGG pathway analyses could offer a more comprehensive view of cell type characteristics. The presentation of scRNAseq results lacks depth in interpretation, particularly regarding the potential roles of different cell types based on their transcriptional profiles and marker genes. Additionally, some figures (2B, C, D, and 7C to H) suffer from information overload and small size, further hampering readability and interpretation.*

Thank you for your valuable suggestions regarding the presentation and interpretation of our KEGG and Gene Ontology (GO) analyses. In response, we revised Figure 2 to enhance clarity and provide deeper insights into cell-type relationships and biological processes.

The revised figure 2 reorganizes GO term analysis into a more intuitive layout, grouping related biological processes and pathways in a structured manner. This approach replaces the dendrogram organization and provides a clearer visualization of key pathways for each cell cluster.

*(4) The pseudotime analysis presented in the study provides modest additional information to what is already manifest from the clustering and UMAP visualization. The central and intermediate transcriptomic profile of cluster 4 relative to other clusters is apparent from the UMAP and the expression of shared marker genes across clusters (as*

*shown in Figure 1D). The statement by the authors that 'the two types of professional phagocytes belong to the same granular cell lineage' (lines 594-596) should be formulated with more caution. While the pseudotime trajectory links macrophage-like (ML) and small granule-like (SGC) cells, this doesn't definitively establish a direct lineage relationship. Such trajectories can result from similarities in gene expression induced by factors other than lineage relationships, such as responses to environmental stimuli or cell cycle states. To conclusively establish this lineage relationship, additional experiments like cell lineage tracing would be necessary, if such tools are available for *C. gigas*.*

We appreciate the reviewer's detailed feedback on the pseudotime analysis and its interpretation. While we acknowledge that the clustering and UMAP visualization provide valuable insights, the pseudotime analysis offers a complementary approach by highlighting significantly expressed genes, including key transcription factors, that might otherwise be overlooked in differential expression analysis based solely on Log2FC between clusters. In our study, the pseudotime analysis revealed transcription factors known to play crucial roles in hemocyte differentiation, providing additional depth to our understanding of hemocyte lineage relationships and functional specialization.

Regarding the statement on lines 594 - 596, we agree that the evidence provided by pseudotime trajectories does not definitively establish a direct lineage relationship between macrophage-like (ML) and small granule-like (SGC) cells. Instead, these trajectories suggest potential developmental connections that warrant further investigation. We propose the following revised sentence (lines 616 to 618) :

"The pseudotime trajectory linking macrophage-like (ML) and small granule-like (SGC) cells suggests a potential developmental relationship within the granular cell lineage; however, this hypothesis requires further validation."

We also concur with the reviewer that additional experiments, such as cell lineage tracing, would be necessary to definitively establish this relationship. Unfortunately, the long-term cultivation of hemocytes in *C. gigas* is currently not feasible. However, we are planning to develop FACS-based approaches to separate the seven hemocyte subtypes, which will allow us to refine their ontology and explore their potential lineage relationships more precisely.

*(6) Given the mention of herpesvirus as a major oyster pathogen, the lack of discussion on genes associated with antiviral immunity is a notable omission. While KEGG pathway analysis associated herpesvirus with cluster 1, the specific genes involved are not elaborated upon.*

Thank you for your valuable observation regarding the lack of discussion on genes associated with antiviral immunity, particularly in the context of herpes virus infection. The KEGG pathway analysis indeed identified a weak signature associated with herpesvirus in Cluster 1, primarily involving genes encoding beta integrins. In humans, beta integrins have been described as receptors facilitating herpesvirus entry (1). However, in the case of naive oysters used in this study, the KEGG signature was subtle, likely reflecting the absence of active viral infection. Additionally, beta integrins are multifunctional molecules that also play critical roles in processes such as cell adhesion, a function attributed to hyalinocytes, as highlighted in our results.

Given the naive status of the oysters and the weak antiviral signature observed, we chose not to discuss these findings in detail in this study. However, ongoing work in our laboratory aims to further investigate the specific hemocyte populations targeted by OsHV-1, which may shed light on the role of integrins in antiviral immunity in oysters.

We hope this clarifies our approach and the context of the KEGG findings. Thank you for bringing this important perspective to our attention.

*(7) The discussion misses an opportunity for comparative analysis with related species. Specifically, a comparison of gene markers and cell populations with *Crassostrea hongkongensis*, could highlight similarities and differences across systems.*

In response to the reviewer's comment, we have added a comparative analysis between *C. hongkongensis* and *C. gigas* hemocyte populations, situating our findings within the broader context of invertebrate immune cell diversity and specialization (lines 747 to 760)

**Reviewer #2 (Recommendations for the authors):**

*(1) Lines 92-93: The authors should add references associated with transcriptomic studies of *C. gigas* hemocytes.*

Thank you for pointing this out. In the revised manuscript, we have added references to previous transcriptomic studies of *C. gigas* hemocytes (line 83).

*(2) Line 121 and 127: The authors should clarify whether 3,000 represents the number of cells loaded or their target for analysis.*

The number of cells processed was optimized to minimize the occurrence of doublets during scRNAseq. Following 10x Genomics Chromium guidelines, we loaded 4,950 cells to successfully recover a target of 3,000 cells, with a doublet rate of 2.4%, well below the target threshold of 2.5%. This information has been added on line 125 of the document. The target was 3,000 cells, and as reported in Supplementary Table S1, the estimated number of cells after STAR-solo alignment was 2,937. This ensures the reliability and accuracy of single-cell transcriptomic data.

*(3) Line 129: "Supp. Table 1" in the text and "Supp. Table S1" in the figure title should be edited.*

The inconsistency between "Supp. Table 1" in the text and "Supp. Table S1" in the figure title has been corrected for uniformity throughout the manuscript (line 134).

*(4) Line 138-139: The authors should clarify that the heatmap displays the top 10 positively enriched marker genes for each cluster, as identified by Seurat's differential expression analysis. It is important to note that the analysis does not explicitly show under-represented transcripts, but rather highlights the contrast between cluster-specific overexpressed genes and their lower expression in other clusters.*

We have clarified that the heatmap displays the top 10 positively enriched marker genes for each cluster, as identified by Seurat's differential expression analysis, and that the analysis highlights cluster-specific overexpressed genes rather than explicitly showing under-represented transcripts (lines 143 - 145).

*(5) Figure 1: The authors should consider improving or potentially removing Figure 1C. The gene IDs are not readable due to their small size, which significantly reduces the informative value of the figure. In addition, the data presented in this heatmap is largely redundant with the more informative and readable dot plot in Figure 1D, which shows both expression levels and the percentage of cells expressing each gene.*



Thank you for your suggestion regarding Figure 1C. In the revised manuscript, we have removed the original panel C from the main figure and transferred it to Supplementary Figure S1K, which improves readability while retaining the relevant data. We have also renumbered the remaining panels for clarity, with the former panel D now designated as panel C. We believe these adjustments address the reviewer's concerns and streamline the presentation of the data.

*(6) Table 1: The authors should clarify in the legend the statistical significance criteria (adjusted p-value) for the genes listed.*

As requested, we have added the adjusted p-value threshold (adj. p-value < 0.05) to the legend of Table 1.

*(7) Line 188: The authors should align the text description of the KEGG pathways in cluster 7 with Figure 2A, describing Wnt signaling pathway and clarifying the terminology "endosome pathway" to ensure consistency.*

In the revised text, we have aligned our description with Figure 2A by explicitly mentioning the Wnt signaling pathway in cluster 7 (lines 193 to 194).

The endo-lysosomal pathway encompasses a series of membrane-bound compartments and trafficking events responsible for the uptake of macromolecules from the extracellular environment, their subsequent sorting in endosomes, and eventual degradation in lysosomes. This pathway is tightly regulated, ensuring not only the breakdown of macromolecules but also the recycling of membrane components and signaling receptors essential for maintaining cellular homeostasis (2). In our study, the KEGG signatures of cluster 7 highlight the involvement of the endo-lysosomal pathway.

*(8) Line 223: The authors should revise the description of cluster 1, avoiding references to morphology at this point in the manuscript, as no morphological data has been presented yet.*

We have revised the description of Cluster 1 to remove references to morphology, ensuring consistency with the data presented at this stage of the manuscript (lines 227 to 229): "Cluster 1, comprising 27.6 % of cells, is characterized by GO-terms related to myosin complex, lamellipodium, membrane and actin cytoskeleton remodelling, as well as phosphotransferase activity."

*(9) Figure 2: The authors should revise Figure 2 to improve the clarity. For Figure 2A, they should address the redundancy in the "Global and overview maps" category by removing overlapping pathways such as carbon metabolism and biosynthesis of amino acids, which are likely represented in more specific metabolic categories (glycolysis, pentose). They could consider grouping similar pathways together, such as combining "Amino acid metabolism" with "Metabolism of other amino acids," and separating metabolic pathways from cellular processes for easier interpretation. They should also address the surprising absence of certain expected pathways like lipid metabolism, nucleotide metabolism, and cofactor/vitamin metabolism, as well as cellular processes such as cell growth and chromatin modeling. Even if these pathways are not enriched in specific clusters, mentioning their absence could provide valuable context for the reader.*

In the revised version of the manuscript, we propose a new representation to facilitate understanding and improve the clarity of the data presentation.

(10) For Figures 2B, C, and D, the authors should significantly increase the font size of text and numbers, ensuring readability at 100% scale in PDF format. They could also add labels directly on each graph to clearly indicate the type of GO terms represented, (Biological Process, Cellular Component, or Molecular Function).

In the revised version of the manuscript, we propose a new representation to facilitate understanding and improve the clarity of the data presentation.

(11) Line 247-250: The authors should revise their description of cell types to follow the same order as presented in Figure 3A.

We have revised the description of cell types in the manuscript to follow the same order as presented in Figure 3A, as requested.

(12) Line 265-266: The authors should develop the significance of the nucleo-cytoplasmic ratio in hemocyte morphology and identification.

We thank the editor for bringing this to our attention and apologize for the discrepancy between the terminology used in the text and the results presented in Figure 3. The text refers to the nuclear-to-cytoplasmic ratio (N/C), while the figure mistakenly displays the inverse ratio, cytoplasmic-to-nuclear ratio (C/N). We recognize that this inversion may cause confusion and will ensure consistency between the text and the figure.

To address this, we propose correcting the figure legend and labels in Figure 3 to align with the terminology used in the text (N/C ratio). This will prevent confusion and maintain clarity throughout the manuscript.

The nuclear-to-cytoplasmic (N:C) ratio, also known as the nucleus:cytoplasm ratio or N/C ratio, is a well-established measurement in cell biology that reflects the relative size of the nucleus to the cytoplasm. This ratio is frequently used as a morphologic feature in the diagnosis of atypia and malignancy in human cells, underscoring its diagnostic value. In the context of our study, we use the N:C ratio to provide a more precise and quantitative description of hemocyte types in *Crassostrea gigas*. Specifically, the N:C ratio allows us to distinguish between different hemocyte morphotypes, such as blasts and granular cells, and to enrich the characterization of their functional specialization. This quantitative measure supports the morphological classification and enhances the reproducibility and clarity of hemocyte identification.

(13) Line 286-294: The authors should review and correct the legend for Figure 3. It seems that the description of results related to Figure 3C has been mistakenly inserted into the legend.

We thank the reviewer for pointing out this issue with the legend of Figure 3. The description of results related to Figure 3C has now been removed from the legend. The revised legend focuses solely on the figure elements, improving clarity and consistency. We believe this adjustment addresses the reviewer's comment effectively.

(14) Figure 3: The authors should revise the legend for Figure 3A to provide more detailed and explicit descriptions of the "Size, shape and particularities" of the ML, SGC, BGC, and VC hemocyte types.

We thank the reviewer for their insightful suggestion to provide more explicit descriptions in the legend for Figure 3A. We have revised the legend to include detailed explanations of the "Size, shape, and particularities" for the ML, SGC, BGC, and VC hemocyte types. Specifically,

we have clarified that size refers to the average granule diameter, shape describes the morphology of the granules (e.g., spherical or elongated), and particularities highlight distinguishing features such as granule color or fluorescence properties observed under specific staining or imaging conditions. We believe this updated legend provides the level of detail requested and enhances the clarity of the figure (lines 294 - 297).

*(15) Figure 4: The authors should clarify the method used for calculating relative gene expression in Figure 4A and Figure 6. They should explicitly state in the figure legend that the expression was normalized to the Cg-rps6 reference gene, as mentioned in line 835. The authors should also provide details on the calculation method used (e.g., 2<sup>-ΔCt</sup> method) and confirm whether the reference gene was expressed at similar levels across all clusters.*

We thank the reviewer for pointing out the need for additional clarity regarding the calculation of relative gene expression in Figures 4A and 6. To address this, we have revised the legends for both figures to explicitly state that gene expression levels were normalized to the reference gene Cg-rps6 and calculated using the 2<sup>-ΔCt</sup> method. We have also confirmed that Cg-rps6 was stably expressed across all hemocyte clusters and explicitly mentioned this in the revised legends. These changes ensure greater transparency and address the reviewer's concerns (lines 342 to 346).

*(16) The authors could consider removing or modifying Figure 4B, as it appears to be redundant with Figure 3C. Both figures show the average percentage of each hemocyte type in the seven Percoll gradient fractions.*

We thank the reviewer for highlighting potential redundancy between Figures 3C and 4B. While both figures present the distribution of hemocyte types across Percoll gradient fractions, Figure 4B serves a distinct and critical purpose in the manuscript. Specifically, it provides the numerical data necessary to understand the correlations shown in Figure 4A, where we analyze the relationship between gene expression levels and the distribution of hemocyte types. These detailed percentages are essential for interpreting the statistical robustness and biological relevance of the correlation matrix, which could not be derived solely from the qualitative visualization in Figure 3C.

*(17) Figure 5: The authors should address the redundancy between Figure S7B and Figure 5B, as they appear to present the same data. In Figure S7B, "SGC" is incorrectly abbreviated as "G".*

In the revised version of the manuscript, we addressed the redundancy between the two figures and we corrected the incorrectly abbreviated SGC.

*(18) Line 412: The authors should correct the typographical error, changing "Pecoll" to "Percoll".*

In the revised version of the manuscript, we correct this typographical error (line 417).

*(19) Line 417: The statement about the inhibitor apocynin likely refers to Figure 5D, not Figure 5C.*

In the revised version of the manuscript, we have corrected this reference error to accurately refer to Figure 5D (line 422).

*(20) Line 441-444: The authors should provide references to support their annotation of cluster 1 as macrophage-like cells based on macrophage-specific genes. These references*

*should cite established literature on known macrophage gene markers, particularly in bivalves or related species if available. They need to clarify whether specific gene markers exist for each of the hemocyte morphotypes they have identified. If such markers are known from previous studies, they should be mentioned and referenced.*

We propose to modify lines 446 to 449 to address the reviewer's concerns. Cluster 1, which we have termed "macrophage-like" due to its pronounced phagocytic activity and reactive oxygen species (ROS) production, is enriched in Angiopoietin-1 receptor expression (Table 1). Angiopoietin receptors belong to the Tie receptor family, which is expressed in a subset of macrophages known as Tie2-expressing monocytes (TEMs) in humans (3–5). While our analysis reveals a strong overexpression of the Angiopoietin-1 receptor, we acknowledge that this receptor is not an exclusive marker for macrophages.

In bivalves, including oysters, no definitive molecular markers have been established for macrophagelike cells as they are defined functionally in this study. Consequently, the identification of such cells relies on their functional characteristics rather than strict marker expression. To clarify, we propose the following revision to the sentence:

Furthermore, this cluster expresses macrophage-related genes, including the macrophage-expressed gene 1 protein (G30226) (Supp. Data S1), along with maturation factors for dual oxidase, an enzyme involved in peroxide formation (Supp. Fig. S8), supporting its designation as macrophage-like based on functional characteristics.

*(21) Figure 7: For Figures 7C to 7H, the authors should increase the font size of gene names and descriptions to ensure legibility in both printed versions and digital formats. To simplify these figures, the authors could consider displaying less differentially expressed genes for each lineage, along with the top genes for each differentiation pathway. If detailed gene information is crucial, they could move the full list to a supplementary table and reference it in the figure legend. Regarding Figure 7I, the authors should reorder the transcription factor genes by cluster and specificity to improve visualization and interpretation, like in Figure 1D.*

Thank you for these valuable suggestions regarding Figure 7. We have revised Figures 7C–H to ensure improved readability. Furthermore, we have simplified these panels by highlighting fewer differentially expressed genes for each lineage. In Figure 7I, we have reordered the transcription factor genes by cluster and specificity, following a layout similar to Figure 1D, to facilitate clearer visualization and interpretation of the data.

*(22) Line 490: The authors should provide more precise references to the specific GO terms and figure panels they are discussing.*

To address this comment, we have revised the sentence and provided additional information in the text to clearly indicate where the corresponding figure panels can be found in the manuscript (line 499)

*(23) Line 510: The authors state that "5 cell lineages could be defined," but the subsequent text and Figure 7C to H actually present 6 distinct lineages.*

We have corrected in the manuscript. 6 lineages could be defined (line 521).

*(24) Line 534: The authors should consider further investigating the pluripotent potential of cluster 4 cells by exploring known or potential stem cell markers in their scRNAseq data.*

Thank you for highlighting the possibility of pluripotent potential of cluster 4. In our current analysis, we did not detect any known stem cell or proliferative markers, nor evidence of a clearly defined hematopoiesis site in the hemolymph. Indeed, previous work suggests that oyster hematopoiesis may occur in tissues such as the gills, implying that stem or progenitor cells might not circulate in the hemolymph under homeostatic conditions. Consequently, it is plausible that our observation of no proliferative cell populations partly reflects their absence in hemolymph, especially in naïve (unstimulated) oysters. To conclusively identify potential progenitor cells and their proliferative activity, further approaches involving deliberate perturbation of hemocyte homeostasis - such as immunological challenge (e.g., Zymosan treatment) combined with lineage-tracing or proliferation assays - would be necessary. These future investigations would not only clarify whether proliferative cells emerge in the hemolymph in response to environmental or pathological stimuli but also help elucidate the broader cellular pathways underlying oyster immune responses.

In response to the reviewer's comment, we have revised the Discussion (lines 695 to 696) and added : "Nevertheless, we did not detect any canonical stem or progenitor cell populations in our dataset, underscoring the need for future investigations - potentially involving immunological challenges and lineage-tracing assays - to clarify whether proliferative cells circulate in the hemolymph or instead reside primarily in tissue compartments."

*(25) Figure S10: The authors should significantly improve the readability of Figure S10 by increasing the font size. Currently, the small font size makes it impossible for readers to discern the information presented.*

Thank you for highlighting the readability concerns regarding Figure S10. In response to your comment, we have increased the overall size and font of the figure, ensuring that all labels and legends are clearly legible in both printed and digital formats. We believe these adjustments will allow readers to more easily interpret the information presented.

*(26) Line 896: The authors should correct the typographical error on line 896 by deleting the additional bracket.*

In the revised version of the manuscript, we correct this typographical error.

*(27) Figure S12: The authors should address the absence of any reference to Figure S12 in the main text of the manuscript.*

The reference to Supp. Figure S12 has been corrected. It was a referencing error between Supp. Figure S11(in the discussion, line 670) and Supp. Figure S12.

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**Reviewer #3 (Public review):**

*The paper addresses pivotal questions concerning the multifaceted functions of oyster hemocytes by integrating single-cell RNA sequencing (scRNA-seq) data with analyses of cell morphology, transcriptional profiles, and immune functions. In addition to investigating granulocyte cells, the study delves into the potential roles of blast and hyalinocyte cells. A key discovery highlighted in this research is the identification of cell types engaged in antimicrobial activities, encompassing processes such as phagocytosis, intracellular copper accumulation, oxidative bursts, and antimicrobial peptide synthesis.*

*A particularly intriguing aspect of the study lies in the exploration of hemocyte lineages, warranting further investigation, such as employing scRNA-seq on embryos at various developmental stages.*

*In the opinion of this reviewer, the discussion should compare and contrast the transcriptome characteristics of hemocytes, particularly granule cells, across the three species of bivalves, aligning with the published scRNA-seq studies in this field to elucidate the uniformities and variances in bivalve hemocytes.*

**Reviewer #3 (Recommendations for the authors):**

*Minor Concerns:*

*(1) In the context of C. gigas, the notable expansion of stress and immune-related genes in its genome stands out. It is anticipated that the article will discuss the expression patterns of classical immune-related genes like TLR and RLR across different cell clusters.*

We appreciate the reviewer's interest in the expression patterns of classical immune-related genes, such as Toll-like receptors (TLRs) and RIG-I-like receptors (RLRs), across different cell clusters in *Crassostrea gigas*. In our single-cell RNA sequencing (scRNA-seq) analysis, we did not detect significant expression of TLR or RLR genes. This absence can be attributed to several factors. First, technical limitations of scRNA-seq: The droplet-based scRNA-seq technology employed in our study captures only a fraction of the transcripts present in each cell approximately 10–20% (<https://kb.10xgenomics.com/hc/en-us/articles/360001539051-What-fraction-of-mRNA-transcripts-are-captured-per-cell>). This inherent limitation often results in the underrepresentation of genes with low expression levels. Consequently, TLRs and RLRs, which may be expressed at low levels in certain hemocytes, could be undetected due to this capture inefficiency. TLRs are typically expressed at low basal levels under resting conditions and are upregulated in response to specific stimuli or pathogenic challenges (1, 2). Given that our study analyzed hemocytes in their basal state, the expression levels of these receptors may have been below the detection threshold of the scRNA-seq platform. Furthermore, as highlighted by De Lorgeril et al. (3) the expression of these immune receptors varies depending on the resistance of the oyster. This variability further underscores the dynamic and context-dependent nature of TLR and RLR expression



To comprehensively assess the expression patterns of TLRs and RLRs across different hemocyte clusters, future studies could incorporate targeted enrichment strategies, such as bulk RNA-seq or single-cell technologies with higher capture efficiencies. Additionally, analyzing hemocytes under stimulated conditions or comparing oysters with varying levels of resistance could provide insights into the inducible and context-specific expression of these immune receptors.

*(2) Clarification is needed in lines 265-266 regarding the nucleo-cytoplasmic ratio (N/C) terminology to prevent confusion, considering the discrepancy with the results presented in Figure 3.*

We thank the editor for bringing this to our attention and apologize for the discrepancy between the terminology used in the text and the results presented in Figure 3. The text refers to the nuclear-to-cytoplasmic ratio (N/C), while the figure mistakenly displays the inverse ratio, cytoplasmic-to-nuclear ratio (C/N). We recognize that this inversion may cause confusion and will ensure consistency between the text and the figure.

To address this, we propose correcting the figure legend and labels in Figure 3 to align with the terminology used in the text (N/C ratio). This will prevent confusion and maintain clarity throughout the manuscript.

*(3) The selection of cluster 4 as the root for pseudotime analysis based on high ribosomal protein expression raises questions. It would be beneficial to elaborate on the inclusion of other genes, such as cell cycle or mitotic-related genes, to validate the pseudotime analysis outcomes.*

We appreciate the reviewer's insightful comment on the significance of ribosomal proteins in stem cell maintenance.

Hematopoietic stem cells (HSCs) are a population of stem cells that are largely cell-cycle-quiescent (G0 phase) with low biosynthetic activity. Upon stimulation and stress HSCs undergo proliferation and differentiation and produce all lineages of hemocytes.

Ribosomal proteins play a multifaceted role in preserving the balance between stem cell quiescence and activation. By ensuring precise regulation of protein synthesis, they allow stem cells to maintain their undifferentiated state while remaining poised for activation when needed. Furthermore, ribosomal proteins contribute to the cellular stress response, safeguarding stem cells from oxidative damage and other stressors that could compromise their functionality. Importantly, ribosomal biogenesis and the dynamic assembly of ribosomes provide a regulatory mechanism that fine-tunes the transition from self-renewal to differentiation, a critical feature of hematopoietic stem cells (HSCs) and other stem cell types. These mechanisms collectively highlight the indispensable role of ribosomal proteins in stem cell biology, underscoring their relevance to our study's findings.

In vertebrate, the maintenance of hematopoietic stem cells (HSCs) and hematopoietic homeostasis is widely acknowledged to rely on the proper regulation of ribosome function and protein synthesis (4). This process necessitates the coordinated expression of numerous genes, including genes that encode ribosomal proteins (RP genes) and those involved in regulating ribosome biogenesis and protein translation. Disruptions or mutations in these critical genes are associated with the development of congenital disorders (5). Among these, Rpl22 (found in cluster 4 with a Log2FC of 1.59) has been shown to play a pivotal role in HSC maintenance by balancing ribosomal protein paralog activity, which is critical for the emergence and function of HSCs (6).

(4) *What is the resolution of the cell clustering employed in the study? Given that cluster 1 potentially encompasses two distinct cell types, Macrophage-Like and Big Granule cells, further sub-clustering efforts and correlation analyses between cluster markers and cell morphologies could aid in their differentiation.*

Thank you for your inquiry regarding the resolution of our cell clustering. As described in the Materials and Methods section, we used the Seurat FindClusters function with a resolution parameter of  $r = 0.1$  for the scRNA-seq dataset. We performed sub-clustering within Cluster 1, resulting in four distinct subclusters. However, despite analyzing various specific markers, we did not identify any marker uniquely associated with the Big Granule Cell (BGC) morphology. Notably, LACC24 specifically marks a subset of cells within Cluster 1, as shown in Supplementary Figure S8, although this gene alone was insufficient to definitively distinguish a distinct BGC population.

(5) *Line 78's statement regarding the primary identification of three hemocyte cell types in *C. gigas*-blast, hyalinocyte, and granulocyte cells would benefit from including references to substantiate this claim.*

We thank Reviewer #1 for their valuable comments, which have allowed us to further improve our manuscript. We have enriched the introduction with the following addition (lines 79 to 82):

“Blast-like cells are considered undifferentiated hemocyte types (Donaghy et al., 2010), hyalinocytes appear to play a key role in wound repair (de la Ballina et al., 2020), and granulocytes are primarily involved in immune surveillance. Among these, granulocytes are regarded as the main immunocompetent hemocyte type (Wang et al., 2017).”

Conclusion:

The authors largely achieved their primary objective of providing a comprehensive characterization of oyster immune cells. They successfully integrated multiple approaches to identify and describe distinct hemocyte types. The correlation of these cell types with specific immune functions represents a significant advancement in understanding oyster immunity. However, certain aspects of their objectives have not been fully achieved. The lineage relationships proposed on the basis of pseudotime analysis, while interesting, require further experimental validation. The potential of antiviral defense mechanisms, an important aspect of oyster immunity, has not been discussed in depth.

This study is likely to have a significant impact on the field of invertebrate immunology, particularly in bivalve research. It provides a new standard for comprehensive immune cell characterization in invertebrates. The identification of specific markers for different hemocyte types will facilitate future research on oyster immunity. The proposed model of hemocyte lineages, while requiring further validation, offers a framework for studying hematopoiesis in bivalves.

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